Upchurch

SAMPLE REPORT



FACILITY CONDITION ASSESSMENT

Mangrove Bay School District

Phase 1 Facility Condition & Energy Assessment —
Representative Portfolio (8 of 41 Facilities) · Sample Report

Cover: Florida school campus. AI-generated illustration for the fictional sample FCA.

PREPARED BY	REPORT DATE	ASSESSMENT PERIOD	SITES ASSESSED
Upchurch — Facility Consulting	October 2026	August – September 2026	8 · 1,197,800 sf gross area

Sample document. Mangrove Bay School District, its facilities, findings, and costs are fictional and were created to illustrate Upchurch's Facility Condition Assessment deliverable. Any resemblance to an actual district or school is coincidental.

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DOCUMENT CONTROL

VERSION	DATA CUTOFF	ASSESSMENT TEAM	STATUS
Rev. 3 · October 2026	September 30, 2026	Dana Whitfield, AIA · Architect Marcus Okafor, PE · Mechanical engineer Elena Ruiz, PE · Electrical engineer	Sample document

REPORT

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SECTION 01

Executive Summary

Upchurch assessed eight Mangrove Bay School District facilities totaling **1,197,800 square feet**, with a current replacement value (CRV) of **$560.9M**. This is a Phase 1 assessment of a representative portfolio; district-wide renewal need will not be known until the remaining 33 facilities are assessed. Assessors inspected every major building system, quantified the major renewal scopes, reviewed four years of utility data, and interviewed site maintenance staff. The result is a single view of what the district's buildings need, when they need it, and which energy and resilience improvements can be delivered alongside that work.

Key figures

CURRENT NEEDS (DEFERRED + DUE BY FY27)

$53.3M

$44/sf

PORTFOLIO FCI

9.5% — Fair

10-YEAR RENEWAL NEED (FY27–FY36)

$209.1M

(37% of CRV) in 2026 dollars; **$243.6M** escalated at 4%/yr

WEIGHTED SITE EUI (FY25)

50.5 kBtu/sf

FY22–FY25 **−3.3% since FY22**



FY22	FY23	FY24	FY25
52.2	51.6	50.9	**50.5**

ENERGY COST (FY25)

$1.70/sf

vs. $1.35 Florida statewide; Gulf Coast districts $1.41–$1.61 (FL DOE, 2024–25)

FY22–FY25 **+9.2% since FY22**



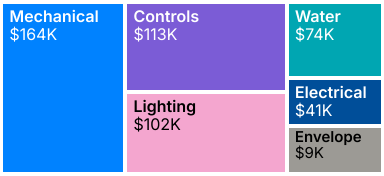
FY22	FY23	FY24	FY25
$1.55	$1.83	$1.72	**$1.70**

PRELIMINARY NET SAVINGS POTENTIAL

$503K/yr

(21% of utility spend), after monitoring fees and solar O&M, before performance-contract M&V

NET SAVINGS BY CATEGORY $/yr



What we found

1. **The portfolio average hides a concentrated problem.** At 9.5% the district rates Fair overall, but four of eight sites rate Poor, and three sites (Mangrove Bay HS, Westshore HS, and Coral Key ES) carry 75% of the $53.3M in current needs.

2. **The near-term load is front-loaded.** FY27 and FY28 together account for $88.9M, or 43% of the 10-year need. After FY27, renewals average $17.3M a year, or 3.1% of CRV, for capital renewal alone. The National Research Council's widely cited guideline is 2–4% of CRV per year for maintenance and repair *combined*, so renewal by itself would absorb most of a typical budget before routine maintenance is funded. Clearing the backlog is a one-time problem; funding the ongoing cycle is a permanent budget decision.

3 **About half of the $53.3M due now is deficiency-driven.** Priority 1 ($7.2M) and Priority 2 ($19.7M) total $26.9M; the rest is age-based renewal. Priority 1 fits within one year of renewal millage. Priority 1 and 2 together consume about 1.8 years of it, before any of the other 33 facilities are funded.

4 **HVAC and the building envelope are 54% of the 10-year need** ($56.6M and $56.1M). On the coast these systems also control humidity, mold risk, and hurricane performance, so their consequence of failure is higher than their dollar share suggests.

5 **Coral Key Elementary warrants a replace-vs-renovate study.** It has the highest FCI in the portfolio (25.3%), a 1962 finger-plan layout, Flood Zone AE exposure, and original building fabric at end of life. Those factors justify the study. The study itself must compare both paths over the same period before the district commits beyond stabilization.

6 **Condition and energy do not move together, so both need measuring.** Tarpon Ridge MS has the best condition in the portfolio (FCI 0.2%) but the third-highest EUI (56.7 kBtu/sf). Its controls are being overridden. A $124K retro-commissioning effort is estimated to save $35K a year (about a 3.6-year payback), the best return in this report.

7 **A performance contract is a delivery tool here, not a funding source.** The HVAC, controls, and lighting package totals $19.8M including audit and development, of which $14.9M is like-for-like equipment replacement the district must fund anyway. Guaranteed savings net of M&V (about $284K a year) support about $3.5M of financing over 20 years: about 72% of the package's efficiency cost and under 2% of the 10-year need. The value is a guaranteed outcome on renewals the district is already funding.

What we recommend

- **Fund Priority 1 items immediately** (8 findings): failed chillers and rooftop units, elevator modernization, and fire sprinkler and accessibility corrections.
- **Retro-commission Tarpon Ridge MS now.** No major equipment replacement is required; fund the $124K commissioning scope. The results will show what to expect from controls work elsewhere.
- **Bundle the FY27–FY28 HVAC and controls renewals into one guaranteed energy performance contract**, funded primarily from capital outlay, with savings covering part of the efficiency increment.
- **Fund envelope, roofing, and flood-resilience work from capital outlay** (1.5-mill levy and, for major renovation, the sales surtax) plus hazard-mitigation grants, and design it to Florida Building Code 9th Edition, which takes effect December 31, 2026.
- **Commission the Coral Key replace-vs-renovate study in FY27.** Proceed with $1.2M of Priority 1, water-quality and life-safety work there, and hold the remaining $6.2M of FY27 renewal scope until the study is complete.

SECTION 02

Board Decisions Requested

The board is asked to act on four items. Together they fund the immediate needs, defer the Coral Key decision to evidence, and set up the FY27–FY28 program.

DECISION	AMOUNT	OWNER	TARGET
1. Approve Priority 1 stabilization funding	$7.2M	Board, on CFO recommendation	Next regular meeting
2. Authorize the Coral Key replace-vs-renovate study; proceed with $1.2M of stabilization (Priority 1, domestic water piping, emergency generator) and hold $6.2M of renewal scope until the study is complete	$1.2M; study fee TBD	Board	Within 30 days
3. Authorize an investment-grade audit for the HVAC and controls performance-contract package; fund Tarpon Ridge retro-commissioning now	$0.6M audit (rolled into the contract if executed; owed by the district if not); $124K RCx	Board / Facilities	Within 30 days
4. Direct staff to return with an FY27–FY28 funding package and a Phase 2 assessment scope for the remaining 33 facilities	—	Superintendent / CFO	Within 90 days

SECTION 03

District Overview & Financial Snapshot

About the district. Mangrove Bay School District serves about **24,600 students** in **41 facilities**: 34 schools and 7 support buildings totaling about 4.15 million square feet along a coastal Gulf county. At the assessed sites' cost per square foot, the district spends an estimated **$7.0M a year on energy** across all facilities. In 2024, voters renewed a half-cent sales surtax for school construction and major renovation through 2034.

Assessment scope. This assessment covers **eight representative facilities** (29% of district floor area), chosen with district staff to span every school type, construction era, and flood exposure. The remaining 33 facilities are recommended for assessment in a second phase (see Next Steps).

Stakeholder engagement. Findings reflect interviews and working sessions with the people who run the buildings:

GROUP	WHO	WHAT THEY CONTRIBUTED
Facilities leadership	Director of Facilities, Maintenance Manager	Capital priorities, deferred-maintenance history, five-year work plan
Trades	HVAC, electrical, roofing, and plumbing supervisors	Failure history, problem equipment, parts availability
Energy & operations	Energy manager, BAS technician	Utility data, schedules, overrides, comfort complaints
Sites	Principals or site leads and head custodians at all eight facilities	Space use, comfort and IAQ issues, operating constraints
Finance	CFO, budget director	Funding sources, millage use, debt capacity
Safety & resilience	Safety & security director, emergency manager	Shelter operations, access control, storm history
Transportation	Transportation director	Fleet facility condition, electrification plans

These same groups weighted and scored the cost–benefit criteria in Section 10.2.

Financial snapshot

Florida districts report spending by function to the Department of Education. Two functions show how much a district invests in its buildings: **Operation of Plant** (utilities, custodial, grounds) and **Maintenance of Plant** (repairs and upkeep).

METRIC (PER UNWEIGHTED FTE STUDENT)	MANGROVE BAY	FLORIDA STATEWIDE (FY2024–25)	DIFFERENCE
General Fund expenditures	$11,240	$11,593	-3%
Operation of Plant (function 7900)	$1,120	$1,045	7%
Maintenance of Plant (function 8100)	$248	$315	**-21%**
Plant O&M share of General Fund	12.2%	11.7%	
Energy cost per square foot (assessed facilities)	$1.70	$1.35	26%

Sources for statewide figures (FY2024–25):

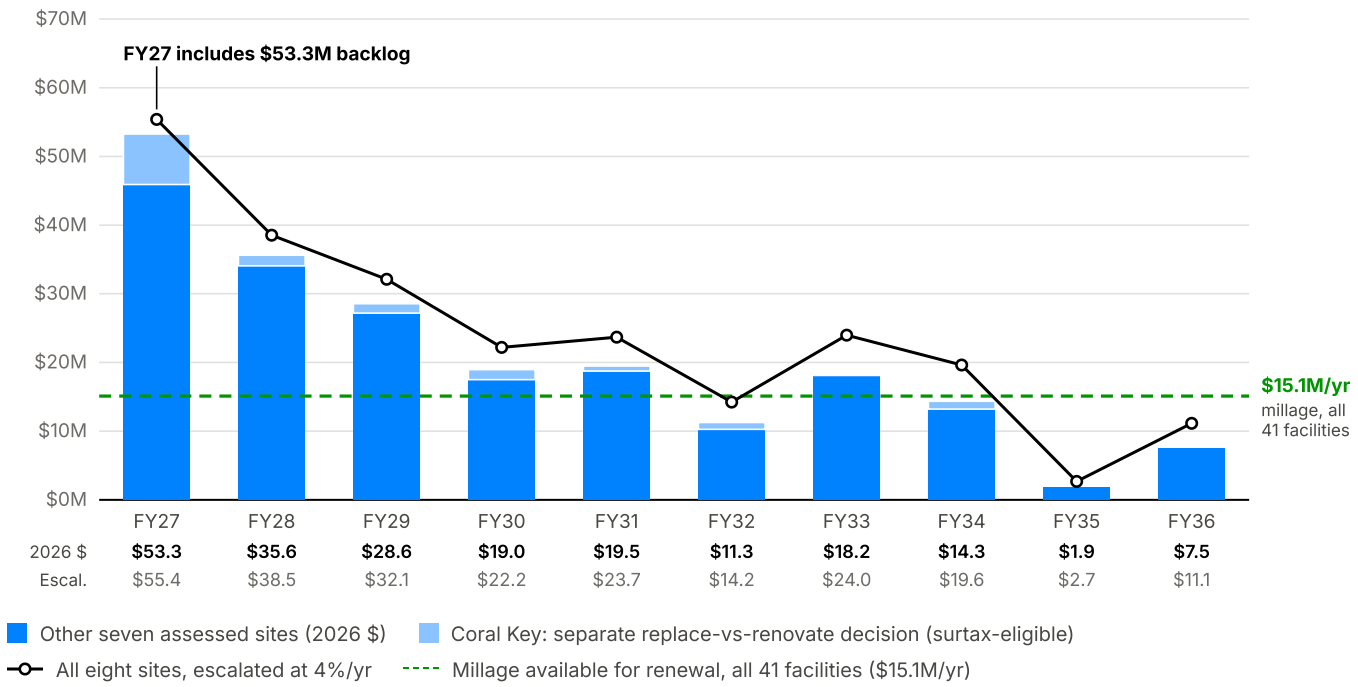
- *General Fund expenditure per student ($11,593): FL DOE, Annual Financial Report Expenditures per Unweighted FTE, General Fund column, State total row (https://www.fldoe.org/file/7507/2425ExpendituresFTE.pdf).*
- *Operation of Plant ($2.886B, function 7900) and Maintenance of Plant ($870M, function 8100): FL DOE School District Annual Financial Reports, "State Cumulative Totals" selection, General Fund expenditures by function (https://www.fldoe.org/finance/fl-edu-finance-program-fefp/school-dis-annual-financial-reports-af.stml). Per-student values divide these by statewide UFTE, which is derived from General Fund total ÷ General Fund per UFTE ($32.01B ÷ $11,593 ≈ 2.76M).*
- *Energy cost per square foot ($1.35): FL DOE, Florida School District Annual Energy Cost Information 2024–2025, State total row, All Energy cost per gross square foot (https://cdn.fldoe.org/file/5599/AnnualEnergyReport2425.pdf).*
- *Mangrove Bay figures are illustrative.*

CAPITAL FUNDING SOURCE	ANNUAL AMOUNT	AVAILABLE FOR RENEWAL OF EXISTING BUILDINGS
Capital outlay millage (1.5 mills)	$43.2M	About 35% ($15.1M) after debt service, buses, technology, and leases
Half-cent sales surtax (through 2034)	$22.0M	Limited by the 2024 referendum to new construction and major renovation
PECO maintenance	Little or none in recent years	Traditional districts have received little PECO maintenance funding recently; not a planning source

What the numbers say. Mangrove Bay spends **more than the state average to operate its buildings and 21% less to maintain them.** That pattern is consistent with a deferred-maintenance cycle, in which utilities and emergency repairs crowd out planned upkeep. Spending ratios alone do not prove it; the condition findings do the work. Four of eight sites rate Poor, and the capital picture points the same way. The eight assessed sites alone need about $17.3M a year in renewals after the FY27 backlog clears. That is more than the roughly $15.1M of millage available each year for renewal across all 41 facilities. The sales surtax can fund Coral Key if the study favors replacement, but as approved by voters it cannot fund routine renewal elsewhere.

FIGURE 1

Renewal need vs. recurring capital capacity, FY27–FY36



The eight assessed sites alone need about $17.3M a year after FY27, more than the renewal millage available to the whole district.

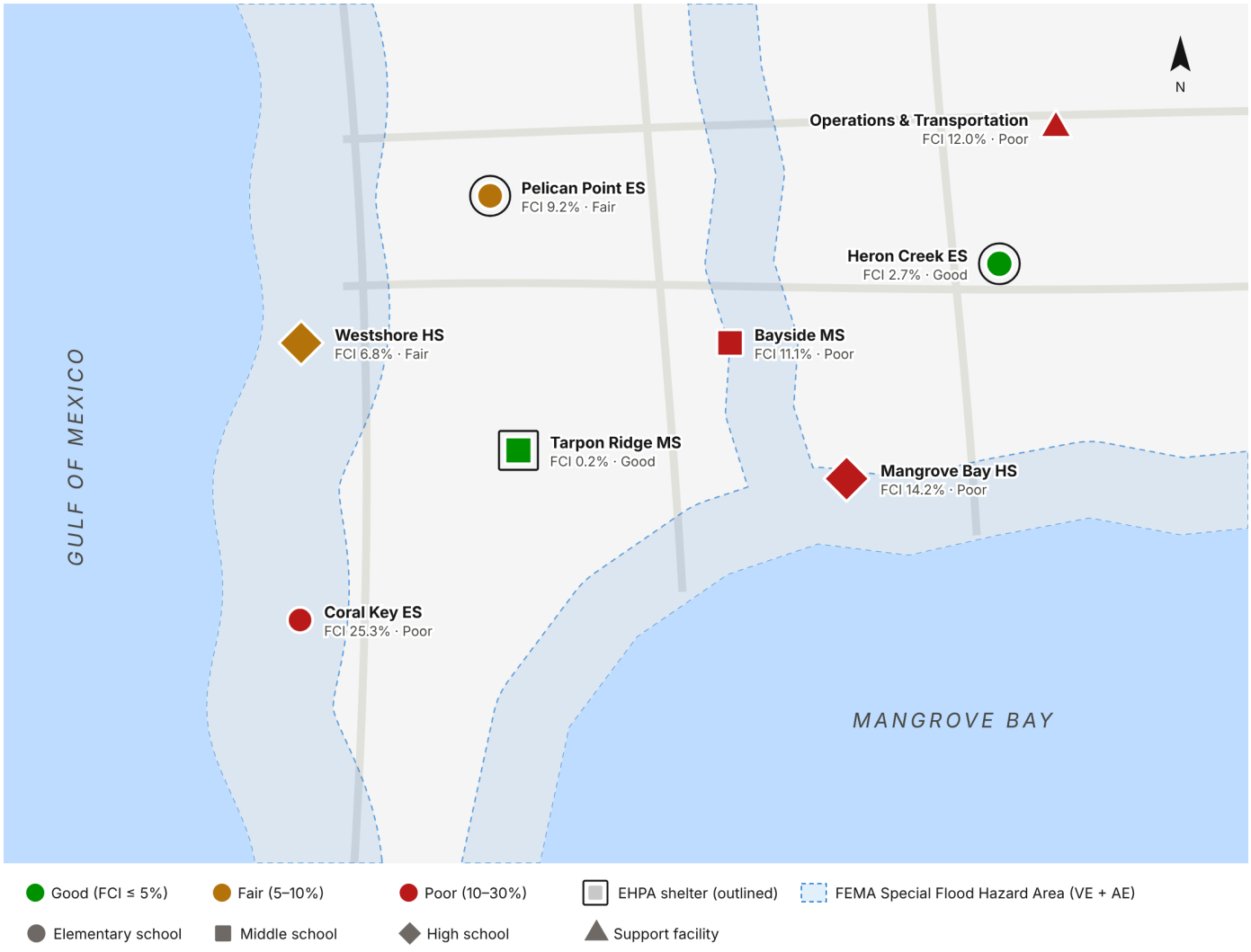
SECTION 04

Portfolio Overview

The assessed portfolio spans seven decades of construction. Four of eight sites sit within one mile of the coast and in FEMA Special Flood Hazard Areas, and three serve as designated Enhanced Hurricane Protection Area (EHPA) shelters.

FIGURE 2

Portfolio sites by current FCI rating



Portfolio sites by current FCI rating. Shape = facility type; pin size = gross square feet; outlined pins are EHPA hurricane shelters. Geography is a schematic of a fictional district; flood zones are detailed in Section 9.

SITE	TYPE	GRADES	BUILT / LAST MAJOR RENO	GSF	ENROLLMENT	FLOOD ZONE	EHPA SHELTER
Coral Key Elementary	Elementary	PK-5	1962 / 1998	68,400	612	AE	No
Pelican Point Elementary	Elementary	K-5	1987 / 2009	84,200	744	X	Yes
Heron Creek Elementary	Elementary	K-5	2006 / —	96,500	868	X	Yes
Bayside Middle School	Middle	6-8	1971 / 1998	142,800	1104	AE	No
Tarpon Ridge Middle School	Middle	6-8	1998 / 2016	158,300	1260	X	Yes
Mangrove Bay High School	High	9-12	1958 / 1994	312,600	2150	AE	No
Westshore High School	High	9-12	1979 / 2003	286,400	1890	VE/AE	No
District Operations & Transportation Center	Support	—	1984 / —	48,600	—	X	No
Total				**1,197,800**	**8,628**		

SECTION 05

Facility Condition Index & 10-Year Renewal Need

The **Facility Condition Index (FCI)** is the cost of a building's current needs divided by its current replacement value (CRV). It is the standard yardstick for comparing buildings of different size, age, and use.

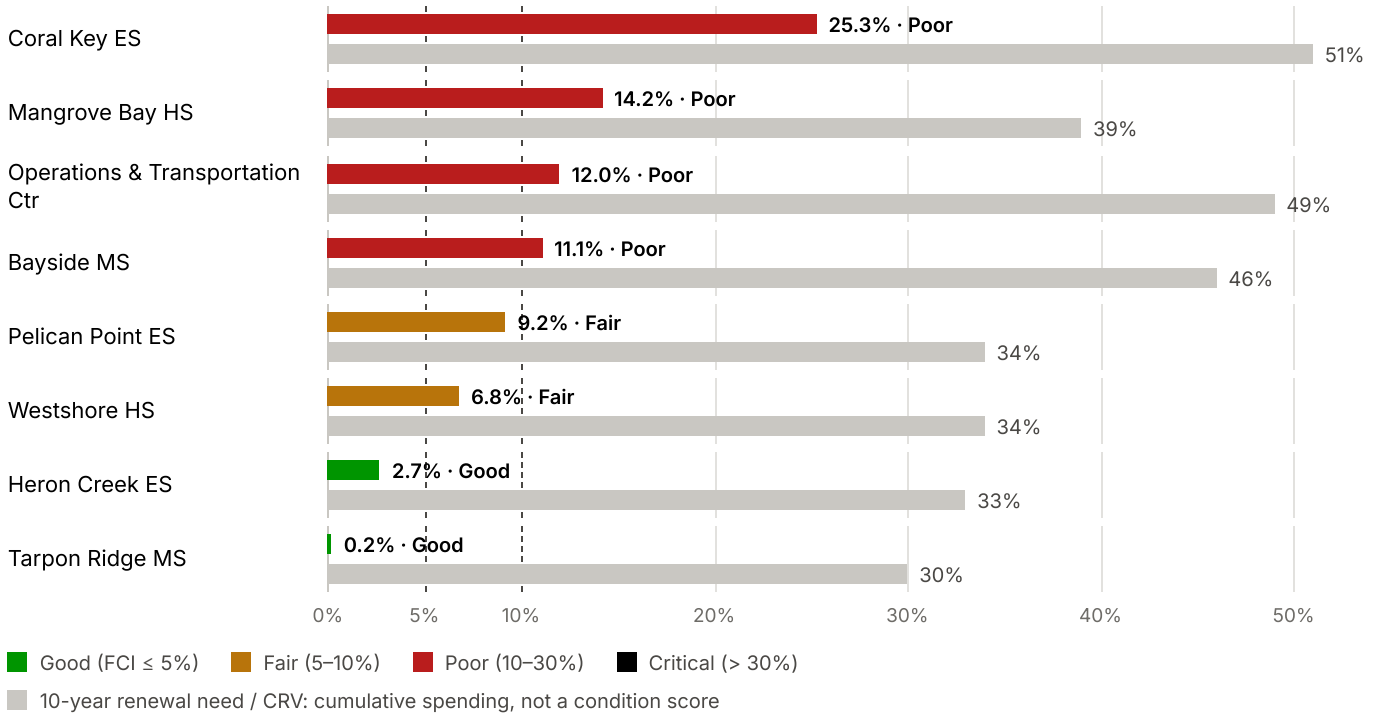
RATING	FCI	WHAT IT MEANS
Good	≤ 5%	Routine maintenance and planned renewals
Fair	5–10%	Renewals coming due; plan funding now
Poor	10–30%	Deferred needs affecting reliability and occupants
Critical	> 30%	Major renovation or replacement should be evaluated

The **10-year renewal need / CRV** ratio is a different measure. It is cumulative spending over FY27–FY36, including normal end-of-life renewals, divided by today's CRV. It sizes the funding requirement. It is **not** a projected condition score: a building that receives this investment on schedule can remain in good condition throughout the decade.

SITE	CRV	CURRENT NEEDS	FCI	RATING	10-YEAR RENEWAL NEED	RENEWAL NEED / CRV
Coral Key Elementary	$29.1M	$7.3M	25.3%	Poor	$14.7M	51%
Mangrove Bay High School	$159.4M	$22.6M	14.2%	Poor	$61.7M	39%
District Operations & Transportation Center	$15.6M	$1.9M	12.0%	Poor	$7.6M	49%
Bayside Middle School	$63.5M	$7.1M	11.1%	Poor	$29.3M	46%
Pelican Point Elementary	$35.8M	$3.3M	9.2%	Fair	$12.0M	34%
Westshore High School	$146.1M	$9.9M	6.8%	Fair	$49.5M	34%
Heron Creek Elementary	$41.0M	$1.1M	2.7%	Good	$13.4M	33%
Tarpon Ridge Middle School	$70.4M	$0.1M	0.2%	Good	$20.9M	30%
Portfolio	**$560.9M**	**$53.3M**	**9.5%**	**Fair**	**$209.1M**	**37%**

FIGURE 3

Current FCI and 10-year renewal need / CRV, by site



Sorted by current FCI, highest first. Dashed reference lines at 5% and 10% FCI. The gray bar is cumulative spending, not a condition score.

Reading the results. Four sites rate Poor today, two rate Fair, and the two newest (Heron Creek ES and Tarpon Ridge MS) rate Good. Coral Key's FCI (25.3%) is nearly three times the portfolio average.

The renewal column shows where money will be needed, not where condition is heading. Heron Creek and Tarpon Ridge each need about a third of their CRV over the decade: Heron Creek's 2006 systems reach end of life together, and Tarpon Ridge's original 1998 air handlers and glazing come due even though its 2016 renovation is in good shape. That is normal lifecycle renewal. If it is funded on time their FCI stays low; if it is deferred, they become the next backlog. The Operations & Transportation Center (49%) is the one to watch: support facilities are routinely deferred in favor of schools, and its renewals are concentrated.

SECTION 06

Energy Use vs. Facility Condition

About the energy data

Utility trends are shown as billed and are not weather-normalized; part of the FY22–FY25 decline may reflect weather. Accounts were reconciled to the eight assessed sites. Savings use FY25 as the baseline. The investment-grade audit will normalize for cooling and heating degree days, occupancy and operating schedules before any savings are guaranteed.

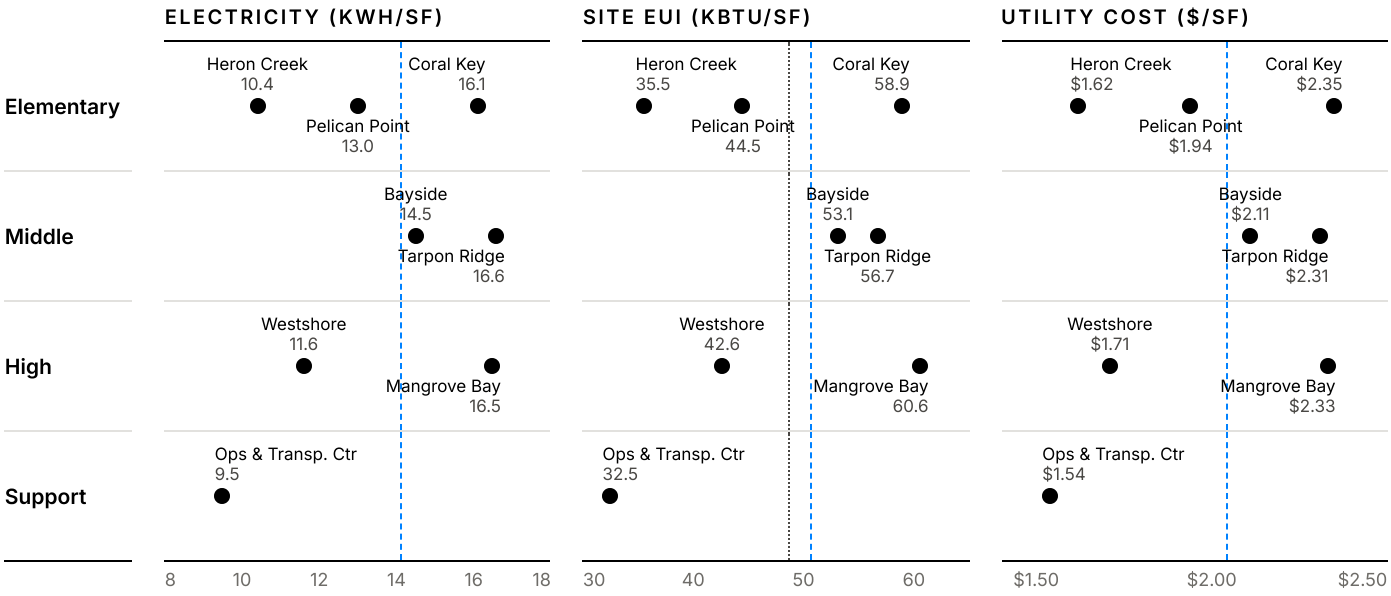
District utility trends (assessed facilities)

UTILITY	FY22	FY23	FY24	FY25	CHANGE FY22–FY25
Electricity (kWh)	17,398,300	17,212,700	16,969,300	16,845,300	-3.2%
Electricity cost	$1,809,420	$2,134,360	$2,019,360	$1,987,750	+9.9%
Blended electric rate ($/kWh)	$0.104	$0.124	$0.119	$0.118	+13.5%
Natural gas (therms)	31,520	30,970	30,190	29,920	-5.1%
Natural gas cost	$51,080	$55,130	$44,670	$43,390	-15.1%
Water & sewer (kgal)	35,130	35,034	35,210	34,884	-0.7%
Water & sewer cost	$365,350	$378,380	$397,870	$411,620	+12.7%
Total utility cost	**$2,225,850**	**$2,567,870**	**$2,461,900**	**$2,442,760**	**+9.7%**
Site EUI (kBtu/sf)	52.2	51.6	50.9	50.5	-3.3%
Utility cost ($/sf)	$1.86	$2.14	$2.06	$2.04	+9.7%

Usage is down; cost is up. Electricity use at the assessed sites fell 3.2% between FY22 and FY25, but total utility cost rose 9.7%. The difference is price: the blended electric rate rose 13.5%, mostly in FY23 when fuel-cost recovery charges spiked, and water and sewer cost rose 13%. The district has little control over rates, but it does control consumption. Each kWh avoided is now worth more than it was three years ago.

FIGURE 4

FY25 utility intensity by school type

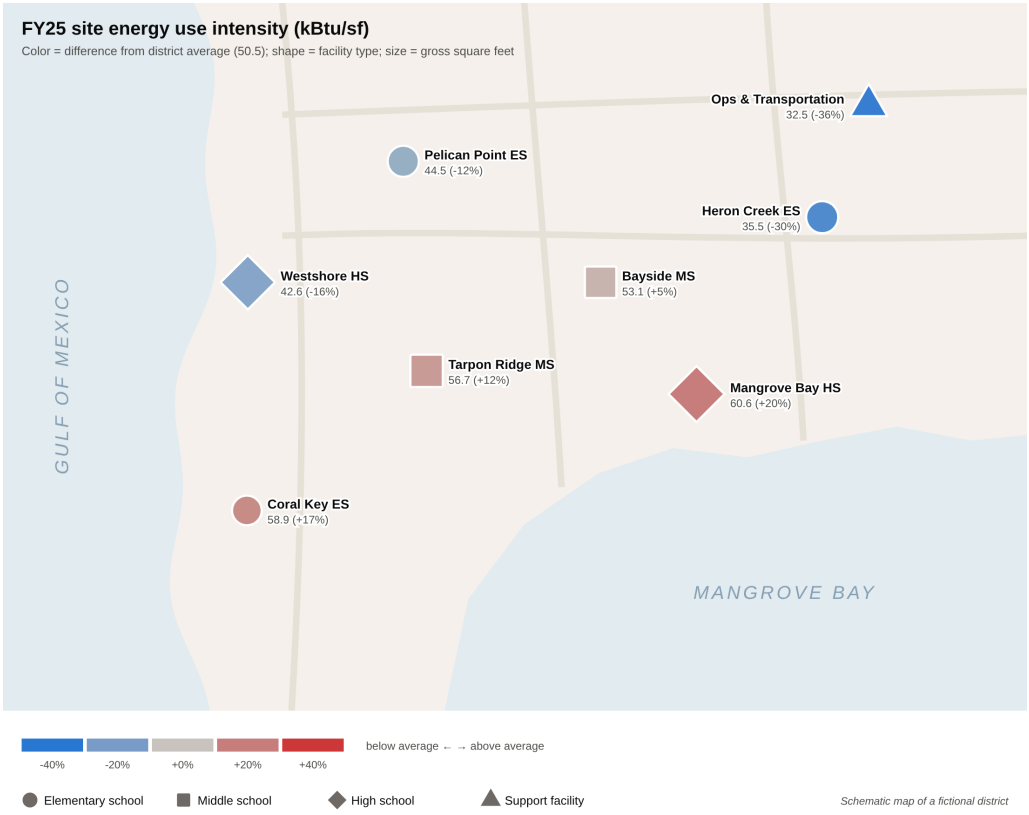


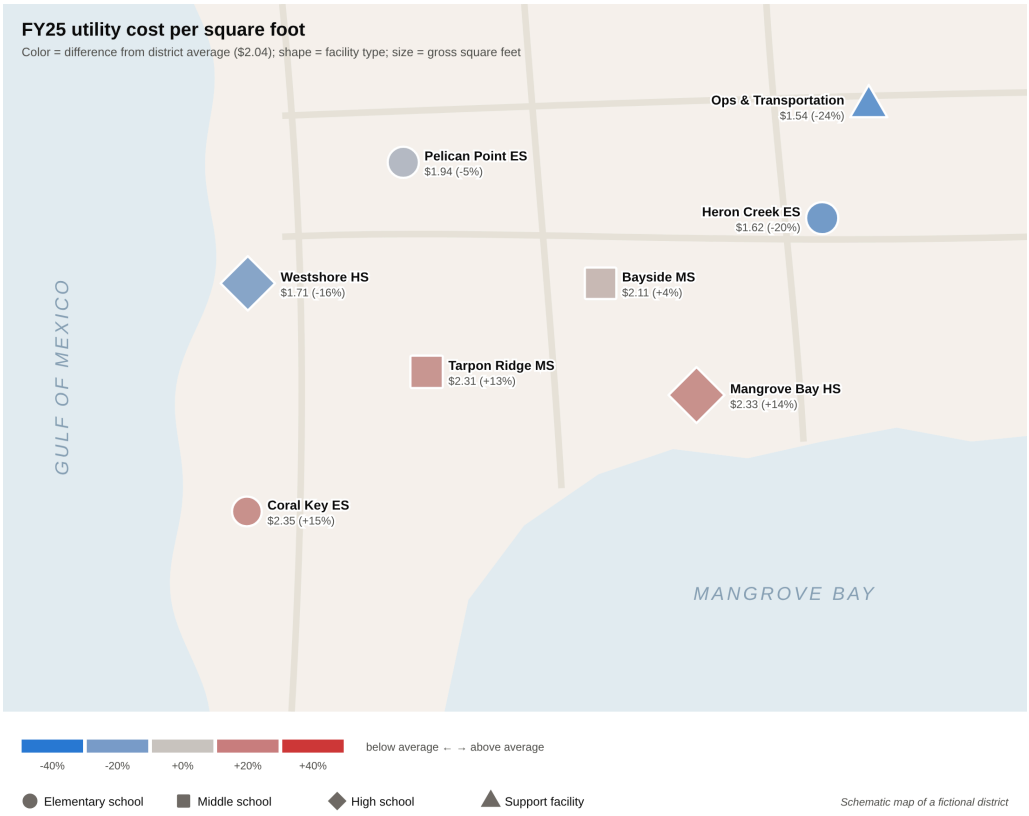
● Site ---- District average (GSF-weighted): 14.1 kWh/sf · 50.5 kBtu/sf · $2.04/sf ····· ENERGY STAR K-12 median 48.5 kBtu/sf

Each dot is a site. Sites with similar values are offset vertically so they don't overlap; vertical position has no meaning.

FIGURE 5

Difference from district average, by site





Color = difference from the district average; shape = facility type; size = gross square feet. Geography is a schematic of a fictional district.

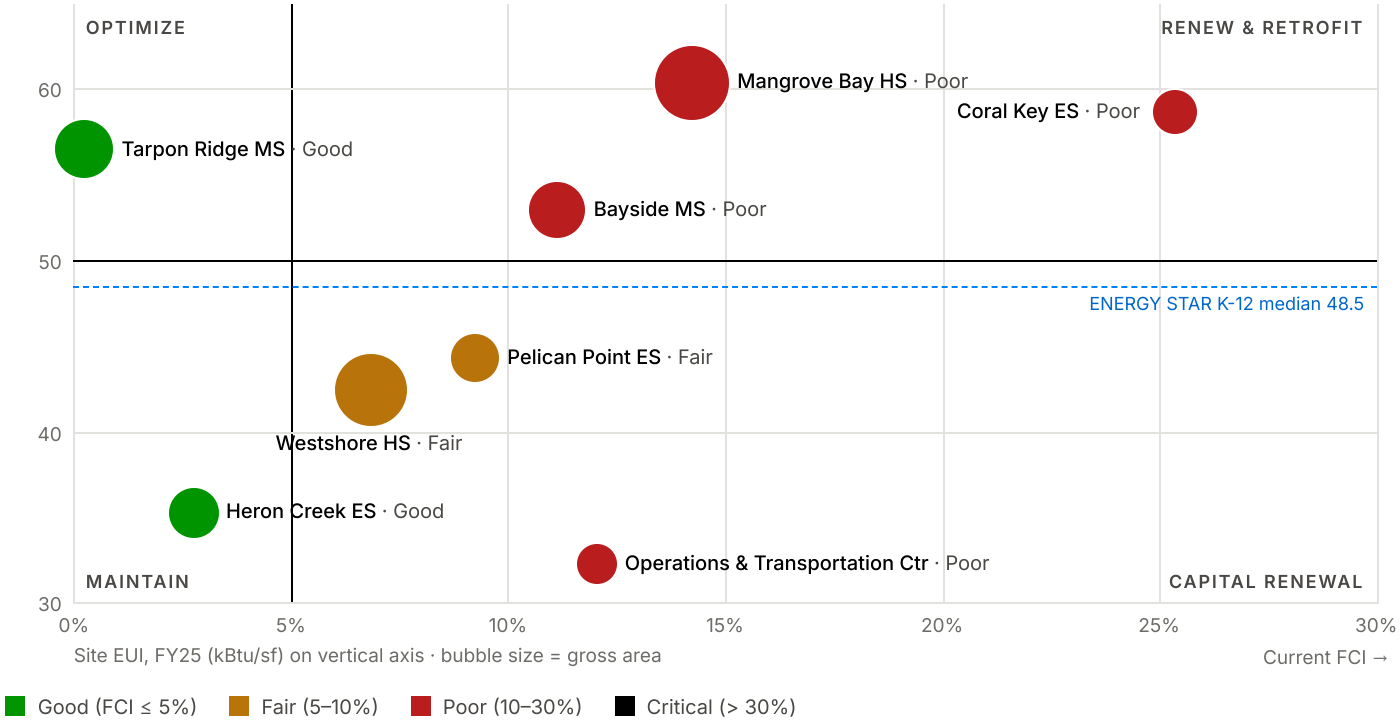
Where the spread is. The highest-EUI elementary school uses 66% more energy per square foot than the lowest, and the spread between the two high schools is 42%. The two middle schools are within 7% of each other, but both sit above the district average. Building age and HVAC condition explain more of the variation than school type does. Coral Key ES ($2.35/sf), Mangrove Bay HS ($2.33/sf), and Tarpon Ridge MS ($2.31/sf) have the highest utility cost per square foot; all three appear on the energy-vs-condition plot below as sites to act on.

Plotting each site's energy use intensity (EUI) against its FCI shows which buildings need capital, which need tuning, and which need both. That determines both the type of project and the funding source.

QUADRANT	SITES	WHAT IT MEANS	PRIMARY PATH
Renew & Retrofit: FCI > 5%, EUI > 50	Coral Key ES, Bayside MS, Mangrove Bay HS	Worn-out systems are wasting energy. Specify every renewal for efficiency.	Capital renewal delivered through a performance contract
Capital Renewal: FCI > 5%, EUI ≤ 50	Pelican Point ES, Westshore HS, Operations & Transportation Ctr	Energy use is reasonable; the condition needs won't be paid for by savings.	Capital outlay; resilience and envelope work
Optimize: FCI ≤ 5%, EUI > 50	Tarpon Ridge MS	Sound equipment with high energy use. Investigate operations before capital.	Retro-commissioning now; capital renewal on its normal schedule
Maintain: FCI ≤ 5%, EUI ≤ 50	Heron Creek ES	Performing reasonably today, but the first major renewal cycle (chillers, controls, lighting) is approaching.	Preventive maintenance; specify upcoming renewals for efficiency

FIGURE 6

Energy use intensity vs. current FCI



Vertical split at 5% FCI; horizontal split at 50 kBtu/sf. Dashed line: ENERGY STAR national K-12 median.

What this means for Mangrove Bay. The district's weighted EUI (50.5 kBtu/sf) is close to the ENERGY STAR national K-12 median of 48.5, but that median is a national reference, not a Florida target. The more telling number is cost: at **$1.70/sf**, Mangrove Bay spends 26% more per square foot on energy than the Florida statewide average ($1.35) and sits above the range of Gulf Coast districts in FL DOE's 2024–25 energy cost report (Charlotte $1.41, Pinellas $1.53, Collier $1.61); it is about 11% above Pinellas. Building-area definitions differ between districts, so the comparison is directional, but this is a district with elevated energy cost, consistent with the three Renew & Retrofit sites, which hold 44% of the floor area and the bulk of the HVAC backlog.

Tarpon Ridge MS is the outlier to act on first. A building in good condition using 57 kBtu/sf points first to controls and scheduling. Field observations support that: 41% of air handlers were in hand, with about 62 hours a week of runtime beyond scheduled occupancy. Prioritize a controls and scheduling investigation, and confirm ventilation rates, actual occupancy and operating hours, and equipment performance before assigning the cause. Retro-commissioning is the vehicle for both. The economizer dampers are repaired under the retro-commissioning scope (TRM-01); replacing the original 1998 air handlers remains a separate FY29 renewal.

SECTION 07

Illustrative data

Operational Service Delivery & Asset Stewardship

THE QUESTION

Are district staff and contractors deployed on the right work, at the right time, and at the right cost to support the facility-condition and capital plan?

7.1 The current service-delivery model

A large district does not simply maintain buildings; it manages a portfolio of in-house trades, specialty contractors, emergency vendors, and capital-project providers. The question is whether each source of capacity is being used for the work it is best suited to perform.

WHAT WE REVIEWED (FY25, JULY 2024–JUNE 2025)

- CMMS work orders, hours, PM records
- Payroll and labor-allocation data
- AP, PO, and contractor invoices
- Maintenance and service contracts
- BAS, utility, and asset evidence
- Facilities, trades, procurement, finance

IN-HOUSE SHARE OF MAINTENANCE LABOR	CONTRACTED SHARE OF MAINTENANCE SPEND	PLANNED VS. REACTIVE / EMERGENCY LABOR	ACTIVE MAINTENANCE & SERVICE CONTRACTORS
81%	**34%**	**32 / 68**	**47**
75,600 of 93,900 hours, incl. est. contractor labor-equivalent	$2.10M of $6.10M	Maintenance hours only; excludes project support and admin	19 under contract, 28 on POs only

FIGURE 7

Who performs the work?

In-house · Contracted · Maintenance and service only; FY25 capital-project contracts ($14.2M) excluded

	LABOR HOURS (CONTRACTOR = EST.)		SPEND		
HVAC	65%	35%	36%	64%	$1.52M
Controls	53%	47%	23%	77%	$0.29M
Electrical	88%	12%	69%	31%	$0.59M
Plumbing	89%	11%	74%	26%	$0.55M
Roofing / envelope	70%	30%	47%	53%	$0.43M
Grounds / custodial support	90%	10%	78%	22%	$1.57M

FIGURE 8

What kind of work is being performed?

Planned PM · Corrective / reactive · Emergency · Capital-project support · Admin / coordination

In-house, all trades	26%	42%	13%	8%	11%
In-house HVAC	21%	49%	19%	6%	5%
Contractor HVAC	34%	44%	22%		

Share of all labor hours. Excluding project support and admin, in-house PM is 32% of maintenance hours.

FINDING

In-house HVAC technicians spend 68% of their hours on reactive and emergency work, while contractors perform not only specialized work and seasonal peaks but also routine PM that recurs every month.

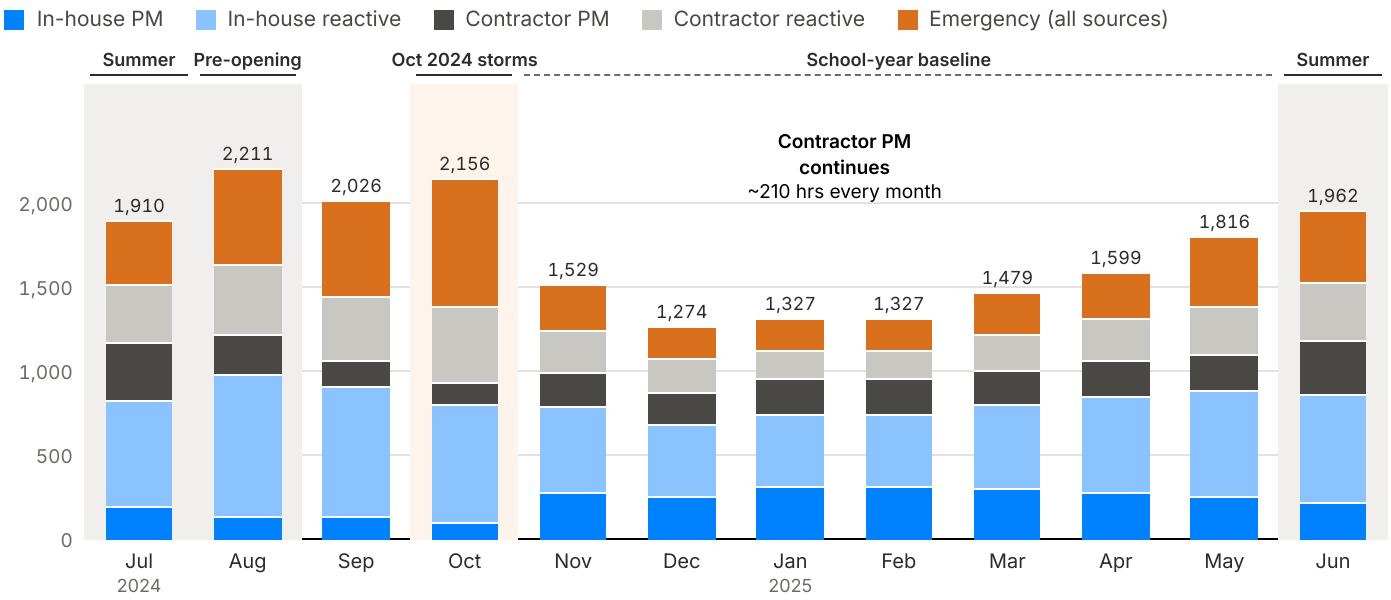
7.2 Baseline capacity versus surge capacity

Illustrative data

Contractors are the right answer for peaks: summer renewals, the weeks before school opens, and storm recovery. They are the wrong answer for steady, predictable work that recurs every month. The chart shows when HVAC maintenance work occurs and who does it.

FIGURE 9

HVAC maintenance labor hours by month, FY25



Capital-project labor is excluded. Contractor hours are estimated labor-equivalent hours (invoice amount ÷ contract labor rate). Invoices also carry materials, markup, and fixed-price work, so spend (Figure 7) is the more reliable comparison.

FRAMEWORK

Right source of capacity for the work

WORK DEMAND	PREFERRED DELIVERY MODEL
Predictable PM, inspections, routine corrective work	**In-house staff**
Specialized scopes or infrequent certifications	**Specialty contractor**
Summer renewals, seasonal peaks, storm recovery, backlog	**Contractors as surge capacity**
After-hours events beyond internal coverage	**Prequalified emergency contractor**
Capital renewal and replacement	**Project contractor with district oversight**

DIAGNOSIS

Contractors are appropriately concentrated in the pre-opening and storm periods, but they also perform about 210 hours of HVAC PM every month from November through April, about 1.4 FTE-equivalents of recurring contractor labor at the 1,800 productive hours a year used for district technicians. That pattern shows the district's in-house HVAC capacity is below its baseline demand, not merely stretched during peaks. Six of the district's nine maintenance vacancies are HVAC positions.

7.3 Contractors: strategic capacity or unmanaged dependence?

Illustrative data

The district paid 47 maintenance and service contractors $2.10M in FY25. $0.55M of that, 26%, was emergency work; in HVAC the emergency share is 31%. Spend is concentrated in HVAC, and so are the governance gaps.

FIGURE 10

Contractor portfolio by service category, FY25

Planned and routine spend Emergency spend

CATEGORY	VENDORS	FY25 SPEND (ALL CONTRACTORS)	EMERGENCY % OF CATEGORY
HVAC / mechanical	9	**$0.97M**	**31%**
Grounds / general	14	**$0.35M**	6%
Roofing / envelope	6	**$0.23M**	**41%**
Controls / BAS	3	**$0.22M**	18%
Electrical	7	**$0.18M**	22%
Plumbing	8	**$0.14M**	**35%**
Total	**47**	**$2.10M · $1.55M planned and routine, $0.55M emergency**	**26% of total**

Three assessed sites (Coral Key ES, Bayside MS, Mangrove Bay HS) account for 46% of assessed-site HVAC contractor spend. All three carry Priority 1 HVAC findings.

CONTRACT COVERAGE

Expirations, overlaps, and gaps

Controls service agreement expired in March 2026 and continues month to month. Two of three BAS platforms are proprietary, so the vendor is effectively single-source.

Overlapping HVAC scopes at four campuses: the PM agreement and the on-call T&M vendor both bill for filter and coil work.

Missing coverage: no contract includes the annual load-bank test on the Pelican Point shelter generator (PPE-02).

Chiller service agreement expires June 2027, in the middle of the FY27–FY28 chiller replacements.

MANAGEMENT VISIBILITY

Can the district see its operating model?

Invoices tied to work orders and assets?	NO	22% of invoices cite a work order; 9% cite an asset.
Contractor cost, response, repeat calls, and quality compared?	NO	No scorecards; response time is not recorded.
Emergency vendors used for true emergencies?	PARTLY	38% of emergency HVAC work orders in the CMMS were on units with overdue PM.
Baseline outsourcing distinguished from surge?	NO	PM and repairs bill on the same POs; we rebuilt the split from invoice text.

Without consistent work-order, invoice, and asset linkage, the district cannot fully see the cost or performance of its operating model. That is itself a capital-planning risk.

7.4 Operational evidence that changes capital decisions

Illustrative data

We matched service-delivery evidence to specific FCA findings. In each case it confirms, accelerates, or qualifies what the capital plan should assume.

FIGURE 11

Service-delivery evidence linked to FCA findings

FCA ASSET / FINDING	SERVICE-DELIVERY EVIDENCE	WHAT IT MEANS FOR THE FCA
CKE-01 Coral Key RTU-4, -7, -11; coils corroded	112 reactive work orders and 31 contractor calls ($48K) in 36 months; no coil-cleaning PM in 24 months	**CONFIRMS** Priority 1 replacement stands; the work history independently flags the same units.
Coastal RTUs Units forecast for extended life at Pelican Point and the Operations Center	HVAC PM 54% on time; the contractor PM scope covers filters, not coil cleaning	**REDUCES CONFIDENCE** The remaining-life extension is at risk; plan for renewal two to three years earlier unless PM coverage changes.
TRM-01 Tarpon Ridge retro-commissioning	41% of air handlers in hand; about 2,300 standing alarms district-wide; controls contract lapsed to month-to-month	**QUALIFIES** The $35K a year savings estimate assumes overrides are released and controls support is in place.
Coral Key roof Priority 2 renewal deferred pending the replace-vs-renovate study	19 emergency leak calls in 36 months, 14 of them August–October	**REQUIRES INTERIM-RISK PLAN** The $0.32M interim restoration allowance should be scheduled before the next storm season.
New capital asset Bayside MS kitchen RTUs, replaced FY24	Not added to the CMMS for nine months; no PM assigned; no operator training at turnover	**RISKS LOST PROJECT VALUE** New equipment without turnover and PM is likely to fail early and void warranty terms.

Source. District CMMS exports, AP invoice detail, contract files, and BAS review (FY23–FY25), matched to deficiency IDs in the Section 12 site reports.

Immediate actions to protect the capital plan

1. Establish interim-risk plans for deferred Priority 1–2 assets.
2. Link CMMS work orders, contractor invoices, and asset records for critical systems and near-term renewals.
3. Define which recurring work is baseline in-house responsibility and which contractor scopes are specialist or surge capacity.

This FCA establishes the current operating baseline and identifies the service-delivery conditions that affect capital-plan reliability. Ongoing performance management, contractor coordination, verification, and capital-plan updates may be addressed through a separate Uptime Ops engagement, with Uptime OS as an optional layer linking asset condition, work history, contractor activity, decisions, and outcomes.

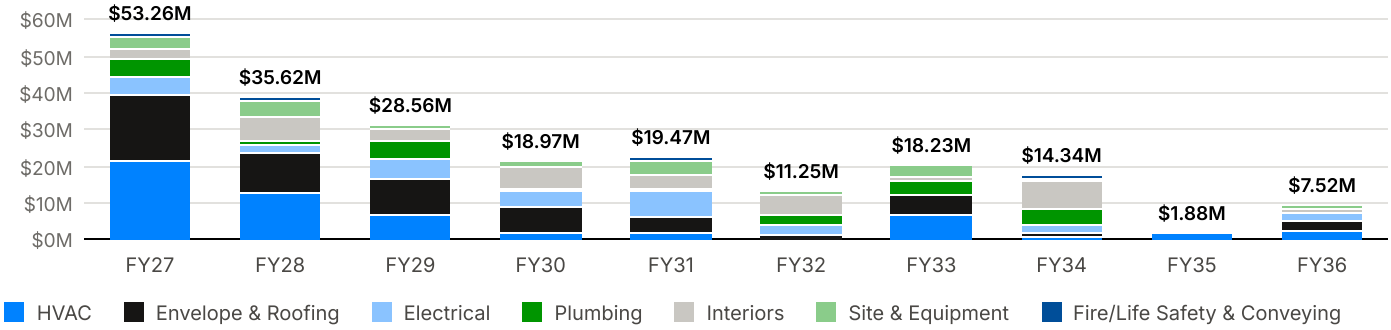
SECTION 08

10-Year Renewal Forecast

The forecast combines each component's remaining useful life, adjusted by field observation, with the deficiencies identified during the assessment. Major scopes (roofs, glazing, HVAC tonnage, outdoor-air capacity, generators, electrical service) are priced from quantities and unit costs. All costs are in 2026 dollars, planning-level (±25%), and include design, contractor general conditions, contingency and soft costs (1.30× construction; Appendix C). They exclude escalation, which the scenario below adds back. Refine through design before budgeting construction contracts.

FIGURE 12

10-year renewal forecast by system group, FY27–FY36 (2026 dollars, $M)



FY27 includes all current (deferred) needs. Totals are labeled above each column.

Escalated funding scenario. The forecast is in constant 2026 dollars so that years and sites compare on the same basis. Budgets are paid in future dollars. Applying 4% annual construction inflation to each year's spending (FY27 is one year after the 2026 cost base) raises the 10-year total from **$209.1M to $243.6M**. The escalated figure is a planning scenario, not a prediction.

PHASE	YEARS	2026 DOLLARS	ESCALATED AT 4%/YR	DIFFERENCE
Phase 1 — Stabilize & Retrofit	FY27–FY28	$88.9M	$93.9M	+$5.0M
Phase 2 — Complete the Envelope	FY29–FY31	$67.0M	$78.0M	+$11.0M
Phase 3 — Sustain	FY32–FY36	$53.2M	$71.7M	+$18.4M
Total	**FY27–FY36**	**$209.1M**	**$243.6M**	**+$34.5M**

Use the 2026-dollar figures to compare sites and set priorities; use the escalated figures for the capital budget and the five-year work program. See Section 10.3 for how the improvement-measure costs relate to this forecast.

SYSTEM	FY27	FY28	FY29	FY30	FY31	FY32–36	10-YEAR TOTAL	SHARE
HVAC	$21.3M	$12.6M	$6.7M	$1.9M	$2.0M	$12.1M	**$56.6M**	27%
Envelope & Roofing	$17.5M	$10.6M	$9.4M	$6.7M	$3.4M	$8.4M	**$56.1M**	27%
Electrical	$4.6M	$1.7M	$4.7M	$3.6M	$6.6M	$5.6M	**$26.7M**	13%
Plumbing	$4.5M	$0.7M	$4.4M	$0.1M	$0.4M	$9.9M	**$19.9M**	10%
Interiors	$2.0M	$5.7M	$2.9M	$5.7M	$2.9M	$12.6M	**$31.8M**	15%
Site & Equipment	$2.8M	$3.9M	$0.5M	$1.0M	$3.7M	$4.4M	**$16.2M**	8%
Fire/Life Safety & Conveying	$0.5M	$0.5M	$0.0M	$0.0M	$0.5M	$0.3M	**$1.7M**	1%
Total	**$53.3M**	**$35.6M**	**$28.6M**	**$19.0M**	**$19.5M**	**$53.2M**	**$209.1M**	**100%**

Observations

- **FY27 is the backlog.** $53.3M is due now. It cannot realistically be funded in one year, so Section 11 sequences it by consequence of failure. Anything not funded rolls into FY28 alongside that year's renewals.
- **HVAC renewals cluster in FY27–FY28** ($33.9M). Replacing chillers, rooftop units, and controls together allows them to be designed as integrated systems and delivered under one performance guarantee.
- **Envelope and roofing follow** ($37.6M in FY27–FY29). These should be designed to Florida Building Code 9th Edition from the start.
- **FY32–FY36 looks light because most aging systems are reset in FY27–FY29.** Their second renewal cycle (controls, finishes, rooftop units) mostly falls after FY36. The next condition update should extend the horizon so those costs aren't mistaken for savings.

Market conditions and timing

Three market forces argue for committing to the FY27–FY28 work rather than letting it slide:

- **Escalation.** At 4% annual construction inflation, every year of deferral adds about 4% to the cost of the work deferred. Deferring the $53.3M backlog by one year costs roughly $2.1M, before counting the emergency repairs that failing equipment generates in the meantime.
- **Long lead times on electrical equipment.** Demand from data centers and utilities has stretched electrical supply chains. Switchgear averaged about 44 weeks to deliver in mid-2025 (Wood Mackenzie), and large transformers far longer. Switchboard, generator, and transfer-switch replacements in this plan (the Operations Center, Pelican Point's shelter generator, and Coral Key's switchboard only if the study favors renovation) should be ordered as soon as each scope is approved. Allow 40–50 weeks from order to delivery when setting construction windows.
- **Refrigerant transition.** Under EPA's AIM Act rules, new commercial rooftop units now use lower-GWP A2L refrigerants (such as R-454B and R-32), and new chillers use low-GWP refrigerants, some of them A2L. Supplies of R-410A for servicing the district's existing fleet will tighten and grow more expensive as the HFC phasedown continues. Equipment still running on R-410A is a growing service-cost risk, which strengthens the case for the Phase 1 HVAC renewals.

After major hurricane seasons, storm-recovery work competes for roofing and electrical contractors across the Gulf Coast. Bidding envelope work in the off-season and packaging it into larger contracts improves both pricing and contractor availability.

SECTION 09

Florida Risk & Resilience Factors

FCI measures what is wearing out. It does not measure what could fail all at once. For a coastal Florida district, four risks deserve their own line in the capital plan.

Salt-air corrosion. Four sites are within one mile of the coast. Rooftop equipment, condenser coils, exterior doors, and generators at these sites carry useful lives 2–3 years shorter than inland equivalents and a 12% cost premium for coated coils, stainless fasteners, and marine-grade finishes. Specifying factory coastal coatings on every new rooftop unit is the cheapest way to extend equipment life at these campuses.

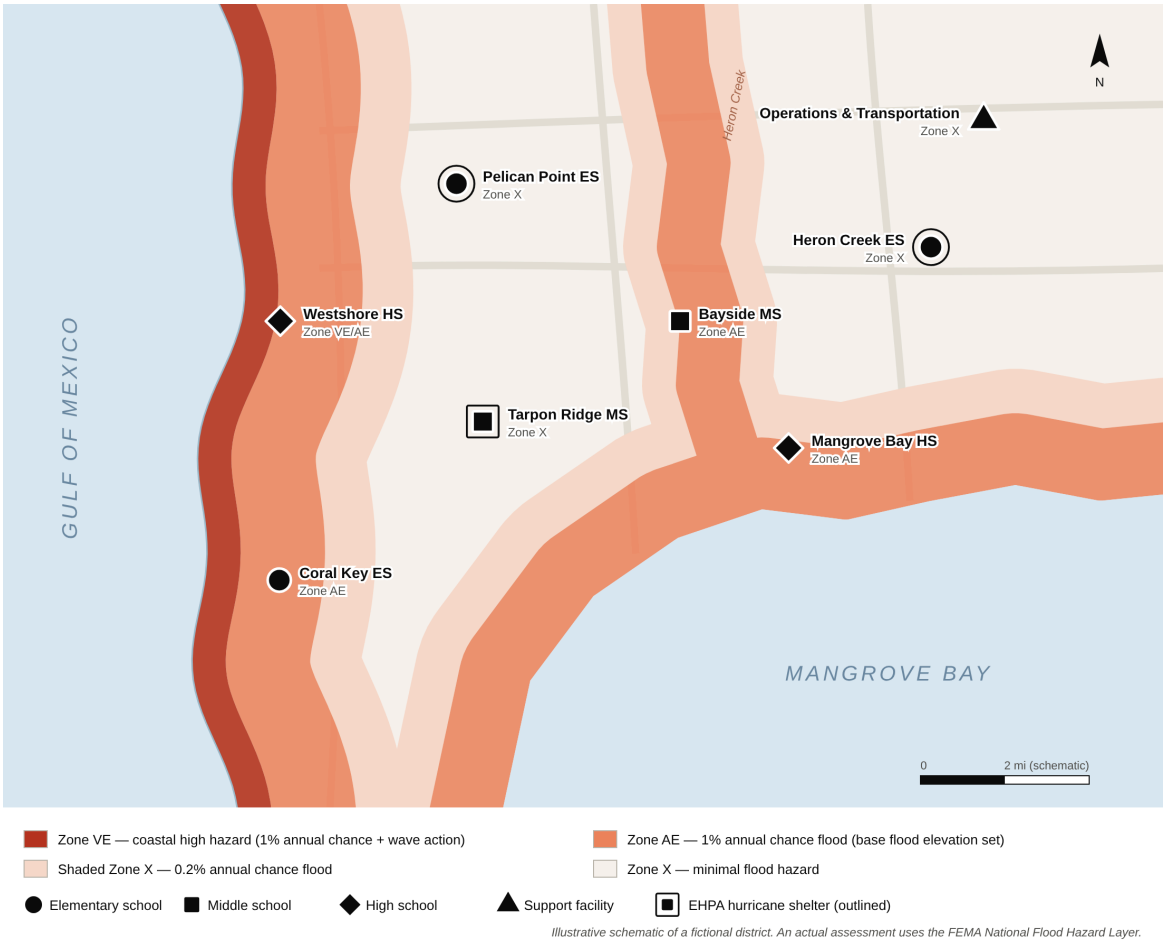
Flood exposure. Bayside MS, Mangrove Bay HS, and Coral Key ES sit in FEMA Zone AE, and Westshore HS sits partly in Zone VE. During the 2024 storm season, surge reached Westshore's ground-floor electrical room. Westshore rates Fair on FCI (6.8%), which understates its risk. Relocating its main electrical service out of the Zone VE wing and above the design flood elevation (screened here at base flood elevation + 1 ft; dry floodproofing is not permitted in Zone VE) ($2.1M) is carried as Priority 2: the difference between a two-day and a two-month recovery.

SITE	FLOOD ZONE	BFE (FT)	LOWEST FINISHED FLOOR (FT)	CRITICAL EQUIPMENT (FT)	FLOOR VS BFE	EQUIPMENT VS BFE + 1	STATUS
Bayside Middle School	AE	8.0	8.6	8.4	+0.6	-0.6	Critical equipment below BFE+1
Coral Key Elementary	AE	10.0	9.2	9.0	-0.8	-2.0	Floor and equipment below flood level
Mangrove Bay High School	AE	9.0	10.4	9.6	+1.4	-0.4	Critical equipment below BFE+1
Westshore High School	VE/AE	13.0	11.5	9.8	-1.5	-4.2	Floor and equipment below flood level
District Operations & Transportation Center	X	—	—	—	—	—	Outside Special Flood Hazard Area
Heron Creek Elementary	X	—	—	—	—	—	Outside Special Flood Hazard Area
Pelican Point Elementary	X	—	—	—	—	—	Outside Special Flood Hazard Area
Tarpon Ridge Middle School	X	—	—	—	—	—	Outside Special Flood Hazard Area

Elevations are illustrative for this fictional district. An actual assessment uses FEMA FIRM panels and elevation certificates or survey data. BFE + 1 ft is a screening assumption; final design elevations depend on the Florida Building Code (ASCE 24 flood design class) and the local floodplain ordinance, which may require more freeboard.

FIGURE 13

Flood zones, school sites, and EHPA shelters



FEMA-style flood zones (VE, AE, shaded X, X); shape = facility type; outlined markers are EHPA shelters. Geography is a schematic of a fictional district.

What the map shows:

- **Westshore HS and Coral Key ES have finished floors below the base flood elevation.** At Westshore, the main electrical room sits 4.2 ft below the BFE + 1 ft screening level, and it is in the Zone VE wing of the building. FEMA does not permit dry floodproofing in Zone VE, so the fix is relocation to an elevated room or the landward (Zone AE) side, carried as Priority 2.
- **Coral Key's flood position affects the replace-vs-renovate decision.** Under the National Flood Insurance Program's substantial-improvement rule, renovation costing 50% or more of a building's market value generally triggers full floodplain compliance (elevating or dry-floodproofing the building). That cost belongs in the study's renovation option.
- **Mangrove Bay HS and Bayside MS sit above the BFE, but their critical equipment does not clear BFE + 1 ft.** Elevating switchgear and generators should be built into those sites' scheduled electrical and HVAC renewals rather than funded as standalone projects.
- **All three EHPA shelters (Pelican Point, Heron Creek, Tarpon Ridge) are outside the Special Flood Hazard Area,** which is where shelters should be. Their resilience gaps are power and envelope, not flooding.

Hurricane shelter readiness. Pelican Point ES, Heron Creek ES, and Tarpon Ridge MS are EHPA shelters. Shelter performance depends on the generator, transfer switch, roof, and glazing all working on the same day. Pelican Point's generator cannot carry full HVAC in the shelter area and is listed for upsizing.

Code transition. Florida Building Code 9th Edition takes effect December 31, 2026. Envelope, roofing, and window projects in this plan should be designed to it from the start.

Recommendation: add a resilience tier to the capital plan, scored on consequence of failure rather than remaining life, so that flood, wind, and shelter items aren't pushed back simply because their effect on FCI is small.

SECTION 10

Facility Improvement Measures & Funding Strategy

Facility improvement measures (FIMs) are where condition and energy meet. Some are renewals the district must make anyway, specified to a higher efficiency standard. Others are pure energy or water measures. The distinction determines how each can be funded.

10.1 Facility improvement measure catalog

The assessment identified **32 facility improvement measures** in eight categories. The catalog is broader than energy: it covers every type of improvement the district could fund, so measures can be compared on the same terms regardless of why they are being considered. Measures marked *need-based* resolve a Priority 1–2 deficiency and proceed regardless of score.

FIGURE 14

Facility improvement measures by category

Mechanical 6

1.1	Packaged rooftop unit replacement (coastal-coated) CKE, PPE, DOC · Phase 1 · CKE-01, PPE-01	Need-based
1.2	Chiller plant replacement & optimization HCE, BMS, MBH, WSH · Phase 1 · BMS-01, MBH-01, HCE-01	Need-based
1.3	Cooling tower refurbishment TRM · Phase 2	
1.4	Dedicated outdoor-air (DOAS) dehumidification CKE, BMS, MBH, TRM · Phase 1 · CKE-02	Need-based
1.5	Air-handler replacement with VAV conversion TRM, BMS, MBH · Phase 2	
1.6	Heat-pump water heaters CKE, PPE, DOC · Phase 3	

Controls & Energy Mgmt 6

2.1	Building automation system upgrade PPE, HCE, BMS, MBH, DOC · Phase 1	
2.2	Retro-commissioning CKE, TRM, WSH · Phase 1 · TRM-01	Need-based
2.3	Monitoring-based commissioning (continuous analytics) CKE, MBH · Phase 1	
2.4	Kitchen hood demand-controlled ventilation BMS, MBH, WSH · Phase 2	
2.5	Utility bill management & tariff review All sites · Phase 1	
2.6	Student & staff energy teams All sites · Phase Optional	

Lighting — 3

3.1 LED fixtures with networked controls (fluorescent sites)
HCE, BMS, MBH · Phase 2

3.2 Exterior & parking-lot LED
All sites · Phase 2

3.3 Athletic field LED
MBH, WSH · Phase Optional

Electrical & Power — 5

4.1 Switchgear & distribution modernization
DOC (CKE pending study) · Phase 2

4.2 EHPA shelter generator upsizing — Need-based
PPE · Phase 1 · PPE-02

4.3 Rooftop solar PV (shelter sites)
PPE, HCE, TRM · Phase Optional

4.4 Battery storage & islanding for shelters
PPE, HCE, TRM · Phase Optional

4.5 Electric school bus charging
DOC · Phase Optional

Envelope — 3

5.1 Roof replacement with insulation upgrade — Need-based
PPE, HCE, BMS, MBH, WSH, DOC · Phase 1–2 · MBH-04, DOC-01

5.2 Impact-rated window replacement — Need-based
BMS, MBH, DOC · Phase 1–2 · BMS-02

5.3 Exterior wall restoration & sealants — Need-based
WSH · Phase 1 · WSH-01

Water & Plumbing — 3

6.1 Fixture retrofits & smart irrigation
All sites · Phase 1

6.2 Domestic water piping replacement — Need-based
CKE, MBH · Phase 1 · CKE-05, MBH-02

6.3 Water submetering & leak detection
All sites · Phase 3

Resilience & Site — 3

7.1 Elevate or relocate critical electrical — Need-based
WSH, MBH · Phase 1 · WSH-02

7.2 Storm drainage & backflow prevention — Need-based
BMS, CKE · Phase 1 · BMS-03

7.3 Wind-rated rooftop equipment anchorage
All coastal sites · Phase 2

Safety & Security — 3

8.1 Access control & secure entry vestibules
All sites · Phase 2

8.2 Fire alarm & mass notification modernization
WSH, DOC, CKE · Phase 1

8.3 Camera system upgrade
All sites · Phase 3

Need-based measures resolve a Priority 1–2 deficiency (listed by ID) and proceed regardless of cost–benefit score.

Where each measure applies. The matrix below shows every measure against every assessed site. Dark cells with a dot are need-based at that site; lighter cells mark sites where the measure applies.

FIGURE 15

Facility improvement measures by site

#	MEASURE	PHASE	CKE	PPE	HCE	BMS	TRM	MBH	WSH	DOC
Mechanical										
1.1	Packaged rooftop unit replacement (coastal-coated)	1	●	●						
1.2	Chiller plant replacement & optimization	1			●	●		●		
1.3	Cooling tower refurbishment	2								
1.4	Dedicated outdoor-air (DOAS) dehumidification	1	●							
1.5	Air-handler replacement with VAV conversion	2								
1.6	Heat-pump water heaters	3								
Controls & Energy Mgmt										
2.1	Building automation system upgrade	1								
2.2	Retro-commissioning	1					●			
2.3	Monitoring-based commissioning (continuous analytics)	1								
2.4	Kitchen hood demand-controlled ventilation	2								
2.5	Utility bill management & tariff review	1								
2.6	Student & staff energy teams	Optional								
Lighting										
3.1	LED fixtures with networked controls (fluorescent sites)	2								
3.2	Exterior & parking-lot LED	2								
3.3	Athletic field LED	Optional								
Electrical & Power										
4.1	Switchgear & distribution modernization	2								
4.2	EHPA shelter generator upsizing	1		●						
4.3	Rooftop solar PV (shelter sites)	Optional								
4.4	Battery storage & islanding for shelters	Optional								
4.5	Electric school bus charging	Optional								
Envelope										
5.1	Roof replacement with insulation upgrade	1–2						●		●
5.2	Impact-rated window replacement	1–2				●				
5.3	Exterior wall restoration & sealants	1							●	
Water & Plumbing										
6.1	Fixture retrofits & smart irrigation	1								
6.2	Domestic water piping replacement	1	●					●		
6.3	Water submetering & leak detection	3								
Resilience & Site										
7.1	Elevate or relocate critical electrical	1							●	
7.2	Storm drainage & backflow prevention	1				●				
7.3	Wind-rated rooftop equipment anchorage	2								
Safety & Security										
8.1	Access control & secure entry vestibules	2								
8.2	Fire alarm & mass notification modernization	1								

#	MEASURE	PHASE	CKE	PPE	HCE	BMS	TRM	MBH	WSH	DOC
8.3	Camera system upgrade	3								

Applies at site · Need-based at site (resolves a Priority 1–2 deficiency) · Colors follow the FIM categories

CKE Coral Key ES · PPE Pelican Point ES · HCE Heron Creek ES · BMS Bayside MS · TRM Tarpon Ridge MS · MBH Mangrove Bay HS · WSH Westshore HS · DOC Operations & Transportation Ctr

10.2 Cost–benefit evaluation

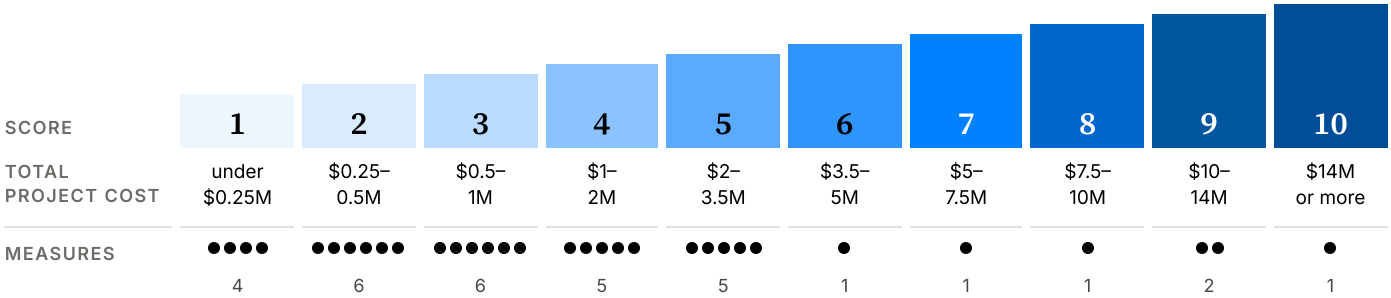
Dollars alone do not capture what an improvement costs a school district or what it returns. Replacing a chiller plant costs capital, but it also costs custodial and maintenance staff time, classroom downtime, and summer schedule. An access-control upgrade saves no energy, but it changes how safely students and staff use the building every day. To compare measures on the terms the district actually cares about, each one was scored on **ten criteria**: five costs and five benefits.

How the scoring works

1. **Weights were set in a workshop** with district facilities, operations, academic, and finance staff. Weights within each side total 100%.
2. **Each measure was scored 1–10 on nine criteria** by the same group, using the definitions below. For cost criteria a higher score means more costly; for benefit criteria a higher score means more beneficial. **The capital-cost score is not a workshop judgment.** It is set from the measure's estimated total project cost across all of its sites, in the fixed dollar bands shown in the figure below. A measure that applies at more sites therefore costs more and scores higher. Each measure's cost and its basis (component renewal, deficiency, savings model, or stated allowance) are on the workbook's Cost-Benefit sheet.
3. **Weighted scores place each measure on a cost–benefit matrix** with four quadrants split at 5. These are judgment scores, not dollars: higher benefit and lower cost scores indicate a more favorable assessment. The difference between the two scores assumes a point of cost and a point of benefit are comparable, and it does not represent financial return.
4. **Need-based measures proceed regardless of quadrant.** A weighted average understates measures that are critical on one dimension (a shelter generator scores 10 on safety and resilience but little elsewhere). The matrix ranks discretionary choices; it does not override deficiencies.

FIGURE 16

Capital-cost score bands



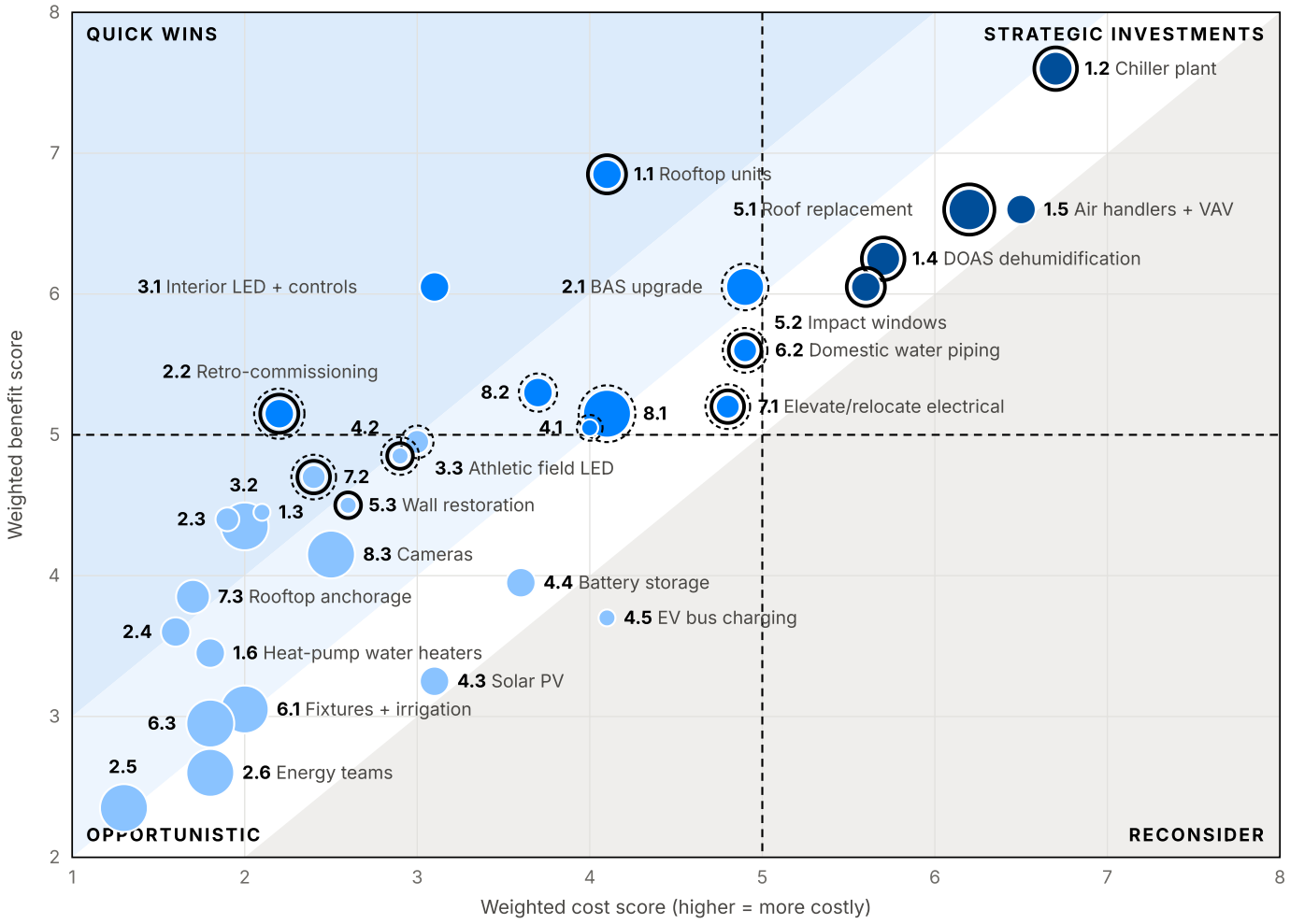
The capital-cost score is set from each measure's estimated total project cost across all of its sites; it is not a workshop judgment. Each dot is one of the 32 measures. Source: workbook Cost-Benefit sheet.

SIDE	CRITERION	WEIGHT	WHAT A 10 MEANS
Cost	Capital cost	40%	Total project cost across the measure's sites of $14M or more (set from the estimate in fixed dollar bands, not scored in the workshop)
Cost	Staff time	20%	Heavy district staff involvement: design reviews, coordination, training
Cost	Downtime & disruption	20%	Classrooms or buildings out of service; work restricted to summer
Cost	Ongoing O&M burden	10%	New systems, skills, subscriptions or maintenance the district must carry
Cost	Implementation risk	10%	Permitting, code-upgrade triggers, phasing or unknown conditions
Benefit	Energy & utility savings	20%	Largest measurable utility savings
Benefit	O&M & reliability	25%	Removes failing equipment, emergency calls and repair spend
Benefit	Student use & learning environment	25%	Comfort, humidity, IAQ, lighting, usable instructional space
Benefit	Staff use & operability	10%	Easier to run, fewer complaints, better tools for custodial/maintenance staff
Benefit	Safety, code & resilience	20%	Life safety, accessibility, code compliance, hurricane/flood resilience

Illustrative sample: weights and scores represent the output of district workshops. In an engagement, Upchurch facilitates these sessions and the district's own weighting drives the result.

FIGURE 17

Cost–benefit score matrix



Quadrant (fill) Quick win · Strategic investment · Opportunistic · Reconsider

Ring Required: resolves a Priority 1–2 deficiency · Borderline: quadrant changes in ≥5 of 41 weight tests

Score difference (benefit – cost) 2 or more · 1 to 2 · 0 to 1 · Below 0

Circle area = number of sites 1 site · 3 sites · 8 sites

Mechanical · **1.1** Rooftop units · **1.2** Chiller plant · **1.3** Cooling tower refurb · **1.4** DOAS dehumidification · **1.5** Air handlers + VAV · **1.6** Heat-pump water heaters

Controls & Energy Mgmt · **2.1** BAS upgrade · **2.2** Retro-commissioning · **2.3** Monitoring-based Cx · **2.4** Kitchen hood DCV · **2.5** Utility bill mgmt · **2.6** Energy teams

Lighting · **3.1** Interior LED + controls · **3.2** Exterior LED · **3.3** Athletic field LED

Electrical & Power · **4.1** Switchgear · **4.2** Shelter generator · **4.3** Solar PV · **4.4** Battery storage · **4.5** EV bus charging

Envelope · **5.1** Roof replacement · **5.2** Impact windows · **5.3** Wall restoration

Water & Plumbing · **6.1** Fixtures + irrigation · **6.2** Domestic water piping · **6.3** Water submetering

Resilience & Site · **7.1** Elevate/relocate electrical · **7.2** Storm drainage · **7.3** Rooftop anchorage

Safety & Security · **8.1** Access control · **8.2** Fire alarm + notification · **8.3** Cameras

Higher benefit and lower cost scores indicate a more favorable workshop assessment. Diagonal bands show the difference between the two scores. They assume a point of cost score and a point of benefit score are comparable, and they do not represent financial return.

Results by quadrant

QUADRANT	MEASURES	WHAT IT MEANS
Quick win (9)	1.1 Packaged rooftop unit replacement (coastal-coated) ●, 2.1 Building automation system upgrade, 2.2 Retro-commissioning ●, 3.1 LED fixtures with networked controls (fluorescent sites), 4.1 Switchgear & distribution modernization, 6.2 Domestic water piping replacement ●, 7.1 Elevate or relocate critical electrical ●, 8.1 Access control & secure entry vestibules, 8.2 Fire alarm & mass notification modernization	Low cost, high benefit. Fund early; most belong in Phase 1.
Strategic investment (5)	1.2 Chiller plant replacement & optimization ●, 1.4 Dedicated outdoor-air (DOAS) dehumidification ●, 1.5 Air-handler replacement with VAV conversion, 5.1 Roof replacement with insulation upgrade ●, 5.2 Impact-rated window replacement ●	High cost, high benefit. The core capital program; sequence by condition and funding.
Opportunistic (18)	1.3 Cooling tower refurbishment, 1.6 Heat-pump water heaters, 2.3 Monitoring-based commissioning (continuous analytics), 2.4 Kitchen hood demand-controlled ventilation, 2.5 Utility bill management & tariff review, 2.6 Student & staff energy teams, 3.2 Exterior & parking-lot LED, 3.3 Athletic field LED, 4.2 EHPA shelter generator upsizing ●, 4.3 Rooftop solar PV (shelter sites), 4.4 Battery storage & islanding for shelters, 4.5 Electric school bus charging, 5.3 Exterior wall restoration & sealants ●, 6.1 Fixture retrofits & smart irrigation, 6.3 Water submetering & leak detection, 7.2 Storm drainage & backflow prevention ●, 7.3 Wind-rated rooftop equipment anchorage, 8.3 Camera system upgrade	Low cost, lower benefit. Do when convenient, bundled with related work or as budget allows.
Reconsider (0)	—	High cost, lower benefit. Defer unless a grant, mandate, or partner changes the math. No measure lands here in this catalog.

● = need-based (required)

What the illustrative scores show

- **The strategic investments are firm.** 1.2 Chiller plant, 1.4 DOAS dehumidification, 1.5 Air-handler VAV conversion, 5.1 Roof replacement and 5.2 Impact windows hold their classification under all or nearly all of the weight tests. Chiller plant replacement (1.2) has both the highest benefit score (7.6) and the highest cost score (6.7). Roof replacement (5.1) is the larger program at $16.8M, but the chiller plant scores higher on staff time, disruption and implementation risk.
- **Only 2 of the 9 quick wins are firm:** 1.1 Rooftop units and 3.1 LED fixtures. The others sit close to a split, and their classification changes when the weights shift (see the sensitivity table below).
- **4 of the quick wins are required corrections:** 1.1 Rooftop units, 2.2 Retro-commissioning, 6.2 Domestic water piping and 7.1 Critical electrical relocation. They proceed on need either way; the score shows they are also reasonable value for what they cost.
- **8.1 Access control and 8.2 Fire alarm modernization score as quick wins without saving energy,** because safety and learning environment carry weight. Both are borderline, so present them as candidates for the district to confirm, not settled results.
- **No measure scores as Reconsider.** 4.5 Bus charging has the lowest score difference in the catalog, and 4.4 Battery storage the next lowest; both remain opportunistic and depend on grant funding.

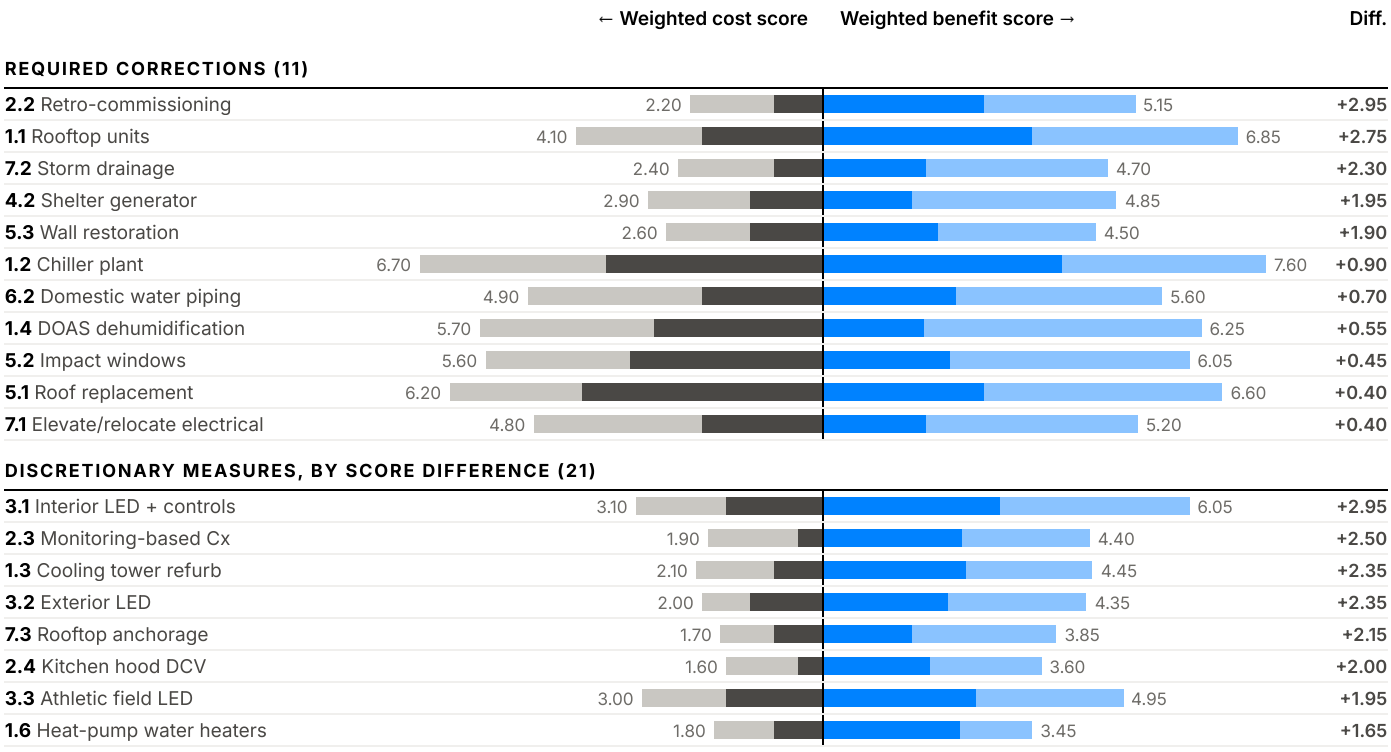
Sensitivity of the classifications. Each of 41 tests moved 10 points of weight between two criteria on the same side (40 tests) or set all weights equal (1 test). These measures change quadrant in at least 5 of the 41 tests:

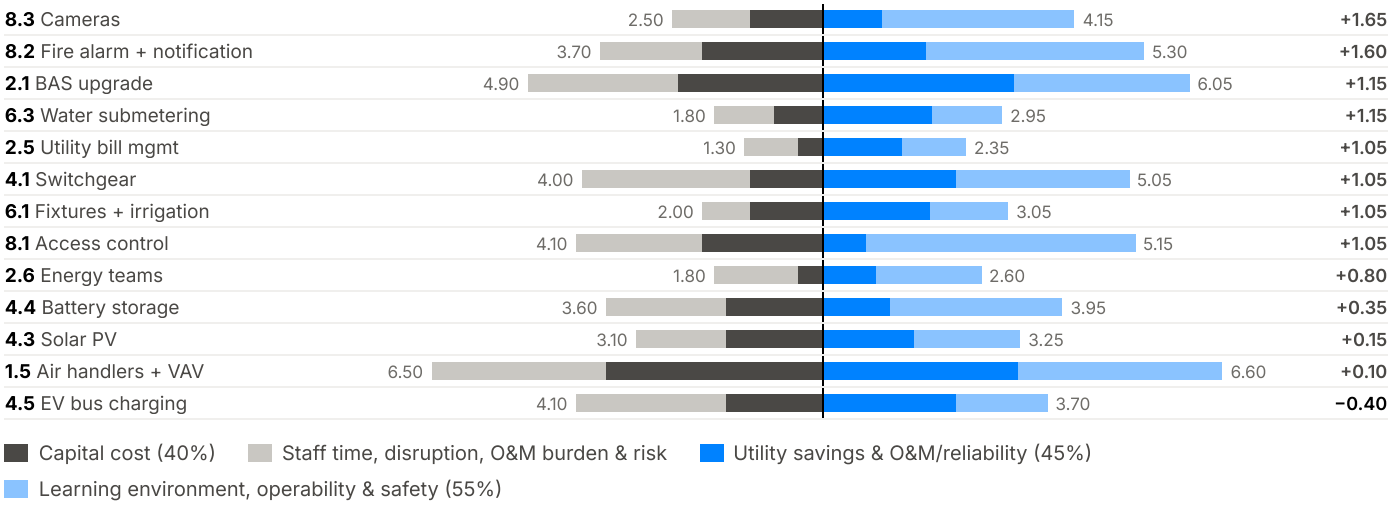
MEASURE	QUADRANT	COST SCORE	BENEFIT SCORE	CHANGES IN	REQUIRED?
7.1 Elevate or relocate critical electrical	Quick win	4.80	5.20	14 of 41	Yes, proceeds regardless
6.2 Domestic water piping replacement	Quick win	4.90	5.60	11 of 41	Yes, proceeds regardless
2.1 Building automation system upgrade	Quick win	4.90	6.05	10 of 41	No
4.1 Switchgear & distribution modernization	Quick win	4.00	5.05	9 of 41	No
8.1 Access control & secure entry vestibules	Quick win	4.10	5.15	8 of 41	No
3.3 Athletic field LED	Opportunistic	3.00	4.95	8 of 41	No
4.2 EHPA shelter generator upsizing	Opportunistic	2.90	4.85	7 of 41	Yes, proceeds regardless
7.2 Storm drainage & backflow prevention	Opportunistic	2.40	4.70	6 of 41	Yes, proceeds regardless
8.2 Fire alarm & mass notification modernization	Quick win	3.70	5.30	6 of 41	No
2.2 Retro-commissioning	Quick win	2.20	5.15	5 of 41	Yes, proceeds regardless

Treat these classifications as provisional until the district confirms its weights in the workshop.

FIGURE 18

What drives the scores





The split shows how the district's weights distribute the scores; it does not measure where economic value comes from.

10.3 Energy savings potential & funding

How savings were estimated. Each site's electricity was split into end uses (cooling, fans, pumps, lighting, plug loads, kitchen, other) using planning assumptions for hot-humid K-12 buildings. Measures were then applied in sequence: controls, then monitoring, then HVAC equipment, lighting, roof, windows, and solar. Each measure acts only on the energy left after the one before it, so no two measures claim the same kWh. Efficiency savings are valued at $0.110/kWh (energy plus partial demand) and solar output at $0.085/kWh (energy charge only). Recurring monitoring fees and solar O&M are deducted to give net savings. M&V for the performance-contract package (3% of guaranteed savings) is a separate cost, carried in the package financing below. **These are preliminary savings potentials for planning; an investment-grade audit is required before contracting.**

#	MEASURE	SITES	INSTALLED COST	EFFICIENCY COST*	NET SAVINGS / YR	PAYBACK ON EFFICIENCY COST	IN PERF. CONTRACT	CURRENT BACKLOG RETIRED	FUTURE RENEWAL ADDRESSED
F1	HVAC replacement - coastal-coated rooftop units (IEER 14.5)	3	$2.2M	$0.3M	$21K	14.0 yrs	Yes	$1.5M	$0.7M
F2	Chiller plant replacement & optimization (high-efficiency plant, VFD pumping)	4	$10.6M	$1.4M	$143K	9.7 yrs	Yes	$6.9M	$3.7M
F3	Building automation system upgrade with commissioning	5	$4.3M	$0.6M	$53K	10.6 yrs	Yes	—	$4.3M
F4	Retro-commissioning (schedules, setpoints, resets); Tarpon Ridge portion delivered outside the contract	3	$0.4M	$0.4M	$54K	7.4 yrs	Partial	—	—
F5	LED fixtures with networked lighting controls	3	$1.7M	$1.7M	$102K	16.9 yrs	Yes	—	—
F6	Roof replacement with insulation upgrade (R-30 ci)	6	$16.8M	$2.2M	$4K	>50 yrs	No	$10.2M	$6.6M
F7	Impact-rated window replacement (low-e insulated glazing)	3	$9.4M	$1.2M	$6K	>50 yrs	No	$2.9M	$6.6M
F8	Water conservation (fixtures, smart irrigation)	8	$0.9M	$0.9M	$74K	12.6 yrs	No	—	—
F9	Rooftop solar PV at EHPA shelter sites	3	$1.1M	$1.1M	$41K	26.6 yrs	No	—	—
F10	Monitoring-based commissioning (continuous analytics)	2	$0.1M	$0.1M	$6K	12.3 yrs	Yes	—	—
	Total		**$47.6M**	**$9.9M**	**$503K**			**$21.5M**	**$21.9M**

*For renewal-driven measures, efficiency cost is a 15% premium over a like-for-like baseline (installed cost × 0.15 ÷ 1.15); payback is calculated on that premium. Roof and window paybacks exceed 50 years because those projects are justified by condition and resilience, with energy as a minor co-benefit. "Current backlog retired" counts renewals already in the FY27 backlog; "future renewal addressed" counts renewals due FY28–FY30 that would be pulled forward.

How the measure costs relate to the renewal forecast. The $47.6M measure total above is not additional to the $209.1M renewal forecast. Most of it is the same money:

ITEM	AMOUNT	IN THE $209.1M FORECAST?
Renewal-driven measures: HVAC equipment, roofs and windows replaced at end of life, priced as high-efficiency equipment	$43.3M	**Yes.** $21.5M is in the FY27 backlog and $21.9M in FY28–FY30 renewals
of which the efficiency premium	$5.7M	**Yes.** The forecast already prices high-efficiency replacements
Tarpon Ridge retro-commissioning	$0.1M	**Yes.** Carried as deficiency TRM-01 in FY27
LED retrofits at HCE, BMS, MBH	$1.7M	**Overlaps.** These sites' lighting renewals ($6.7M, FY29–FY32) are in the forecast; the retrofit pulls the fixture share forward, so reduce those renewals when it proceeds
Other standalone measures: retro-commissioning at Coral Key and Westshore ($0.3M), water ($0.9M), solar PV ($1.1M), monitoring ($0.1M)	$2.4M	**No.** Additional
Investment-grade audit and development (performance-contract package)	$0.6M	**No.** Additional
Coral Key interim roof-restoration allowance	$0.3M	**No.** Additional
Coral Key replace-vs-renovate study	Not priced	**No**
Coral Key replacement school, if the study favors it	Not priced (CRV $29.1M plus flood compliance and site work)	**No.** Would replace Coral Key's $14.7M renewal need, not add to it
Escalation	+$34.5M at 4%/yr	**No.** See the scenario in Section 8
The other 33 district facilities	Not assessed	**No**
Recurring costs: monitoring fees, solar O&M, performance-contract M&V	Annual	**No.** Operating budget; netted from savings

Bottom line: in 2026 dollars, the assessed program costs about **$212.4M**: the $209.1M forecast plus $3.3M of additions. That is before escalation and before any Coral Key replacement decision.

Notes on specific measures

- **LED lighting (F5)** is recommended only at the three sites still on fluorescent fixtures, and even there paybacks run longer than commercial buildings see because schools operate roughly 2,600 hours a year. The other sites completed LED tube retrofits between 2014 and 2019; full fixture replacement there is a renovation item, not a standalone energy measure.
- **Solar PV (F9)** is sized for the shelter sites (400 kWdc total, about 580 MWh/yr at 1,450 kWh/kWdc). At current tariffs the payback is long, so it is excluded from the performance contract. **PV alone does not keep a shelter powered during a grid outage.** That requires battery storage and islanding controls, which are not included in this cost and should be pursued through resilience grant funding. With the federal elective-pay credit (30%), payback falls to about 19 years, but the array must be in service by December 31, 2027. Decide in FY27 or treat solar as deferred.

- **Chiller plants (F2)** are credited by plant type: water-cooled plants from roughly 1.0 to 0.66–0.68 kW/ton, and Heron Creek's air-cooled plant from 1.35 to 1.05 kW/ton (a like-for-like air-cooled replacement; converting to water-cooled is not assumed).
- **Monitoring-based commissioning (F10)** is credited only for direct fault-detection savings, and is recommended only where those savings, net of the subscription fee, repay the implementation cost within 15 years (2 of 8 sites). Elsewhere it is a persistence and operations tool to be justified on those terms, not on energy savings. Its larger value, keeping controls savings from drifting back, is not monetized here.

Performance-contract package (F.S. 1013.23)

COST ELEMENT	AMOUNT
Base renewal cost (like-for-like replacement the district must fund regardless)	$14.9M
Efficiency premiums on renewals	$2.2M
Standalone energy measures (RCx at Coral Key and Westshore, LED, monitoring)	$2.1M
Investment-grade audit & project development (3%)	$0.6M
Total project cost	**$19.8M**
Efficiency cost (premiums + standalone measures + development)	$4.9M
Estimated net annual savings (after monitoring fees)	$344K
Guaranteed annual savings (85% of estimate)	$292K
M&V (3% of guaranteed savings)	$9K
Guaranteed savings net of M&V	$284K
Savings-supported financing (PV, 20 yrs at 5%)	**$3.5M**
Capacity as share of efficiency cost	72%
Capital funding still required (total project – financing)	**$16.3M**

Tarpon Ridge retro-commissioning ($124K, about $35K/yr) proceeds immediately as a standalone project and is excluded from the contract baseline.

What this means. Financing is sized on guaranteed savings, not the estimate, because guarantees are typically set below projected savings and controls savings degrade without ongoing commissioning. On that basis, savings carry about 72% of the package's efficiency increment and none of the underlying renewal cost. The case for a performance contract here is a single accountable contractor, a guaranteed outcome, and measured savings on HVAC replacements the district is already funding. It does not create new capital. Any proposal claiming energy savings will fund the district's backlog should be met with skepticism. Savings potential excludes avoided emergency repairs and O&M savings, which would strengthen the case but are rarely guaranteed.

Funding sources available to a Florida district

SOURCE	WHAT IT FUNDS	NOTES FOR MANGROVE BAY
Capital outlay millage (up to 1.5 mills, s. 1011.71(2), F.S.)	Maintenance, renovation, repair, equipment, leases	The primary source for building renewal; currently about $15M/yr available after other commitments
School capital outlay sales surtax (s. 212.055(6), F.S.)	Construction, reconstruction, or improvement of school facilities, as defined by the referendum	Mangrove Bay's 2024 referendum limits use to new construction and major renovation; a natural fit for Coral Key
PECO (Public Education Capital Outlay)	Maintenance and renovation allocation	Traditional districts have received little or none in recent years; do not plan on it
General obligation bonds (voter referendum)	Large capital programs	Option if the backlog plus Coral Key exceeds pay-as-you-go capacity
Certificates of participation (lease-purchase)	Capital projects without a referendum	Repaid from millage; reduces future millage available for renewal
Guaranteed energy performance contract (s. 1013.23, F.S.)	Energy and water measures, delivered with renewals	Section 10.3: savings carry about 70% of the efficiency cost, not the renewal cost
Utility rebates (FEECA-approved programs)	Lighting, HVAC, controls	Duke Energy Florida and other utilities run commercial rebate programs; apply before purchase
FEMA Hazard Mitigation Grant Program (through FDEM)	Flood mitigation (including elevation or relocation of critical equipment), wind retrofits, shelter generators	Funded after presidential disaster declarations; the 2024 storms opened HMGP funding in Florida. Federal processing has slowed, so allow long timelines
FEMA BRIC	Pre-disaster mitigation	FEMA moved to end BRIC in April 2025 and a federal court ordered it restored in December 2025. FEMA then opened a $1 billion FY2024–25 funding round on March 25, 2026 (applications closed July 23, 2026; no applicant may receive more than 15% of the total). Awards are competitive and flow through the state, so keep BRIC out of committed funding until an award is made, and have the Westshore electrical relocation scoped and costed for the next round
Resilient Florida (FDEP)	Flood and sea-level-rise adaptation	Counties and municipalities apply, not school districts; pursue the Westshore electrical relocation/elevation through a county partnership
Federal clean energy credits (elective pay)	Battery storage; solar on a tight timeline	The begin-construction window closed July 4, 2026, so new solar must be placed in service by December 31, 2027 (2025 federal tax law). Storage credits continue. Confirm with tax counsel

SECTION 11

Prioritized Action Plan

Every deficiency is assigned a priority based on consequence of failure, then sequenced into phases that match how districts fund work.

PRIORITY	DEFINITION	BUDGET YEAR
1 — Immediate	Life safety, code, failed systems, active damage	FY27
2 — Near-term	Poor condition, likely to fail within two years; resilience gaps	FY27–FY28
3 — Planned	Renewals due within five years	FY29–FY31
4 — Deferred	Monitor; renew at end of life	FY32–FY36

11.1 Priority 1 — Immediate (8 findings)

ID	SITE	SYSTEM	FINDING	SCOPE	EST. PROJECT COST	IN 10-YR FORECAST
CKE-01	Coral Key ES	HVAC	RTU-4, -7 and -11 condenser coils fully corroded; units tripping on high head pressure.	3 of 14 RTUs (approx. 33 of 155 tons)	$189,000	Yes
CKE-02	Coral Key ES	HVAC	Classrooms 104-118 above 78°F / 65% RH during walk-through; the 100-wing (8 classrooms, ~7,200 sf) has no dedicated outdoor-air dehumidification.	100-wing DOAS only, ~3,000 cfm	$151,000	Yes
CKE-07	Coral Key ES	Accessibility	Accessible route from bus loop to main entry has 4.1% cross-slope and a 1.5" vertical change at sidewalk joint.	Repair — included in FY27 needs (not a lifecycle component)	$49,000	Yes
CKE-08	Coral Key ES	Fire Protection	Sprinkler heads in kitchen hood area painted over; 5-year internal pipe inspection overdue.	Repair — included in FY27 needs (not a lifecycle component)	$16,000	Yes
PPE-01	Pelican Point ES	HVAC	9 of 22 RTUs with corroded condenser coils and active refrigerant leaks.	9 of 22 RTUs	$529,000	Yes
BMS-01	Bayside MS	HVAC	Chiller CH-2 (2002) out of service; school running on single chiller with no redundancy.	Full component renewal	$1,828,000	Yes
MBH-01	Mangrove Bay HS	HVAC	Central plant chillers (2003) near failure; tower basin leaking; no N+1.	Full component renewal	$3,999,000	Yes
MBH-03	Mangrove Bay HS	Conveying	Elevator #2 (1994) out of service 3x this school year; hydraulic leak.	Full component renewal	$481,000	Yes
Total					**$7.2M**	

11.2 Priority 2 — Near-term (14 findings)

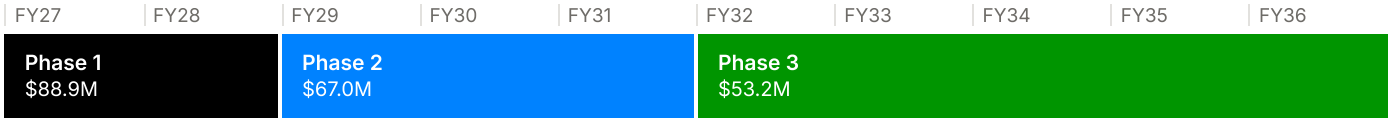
ID	SITE	SYSTEM	FINDING	SCOPE	EST. PROJECT COST	IN 10-YR FORECAST
CKE-03	Coral Key ES	Exterior Enclosure	Original 1962 single-pane steel windows in 100/200 wings; non-impact glazing, failed glazing seals and water intrusion at sills.	100/200 wings, approx. 75% of glazing	$859,000	Yes — on hold
CKE-04	Coral Key ES	Roofing	Modified bitumen roof (2003) at end of life; ponding over cafeteria, 14 patched leaks on work-order history.	Full component renewal	$2,367,000	Yes — on hold
CKE-05	Coral Key ES	Plumbing	Original galvanized domestic water piping; low pressure and rust-colored water at far fixtures.	Full component renewal	$598,000	Yes
PPE-02	Pelican Point ES	Electrical	EHPA shelter generator (2009) undersized for full HVAC in shelter area.	Full component renewal	$525,000	Yes
HCE-01	Heron Creek ES	HVAC	Air-cooled chillers at 20 yrs with condenser coil corrosion.	Full component renewal	$1,101,000	Yes
BMS-02	Bayside MS	Exterior Enclosure	1971 windows, non-impact; storm shutters stored off-site.	Full component renewal	$2,870,000	Yes
BMS-03	Bayside MS	Sitework	Storm drainage backing up at courtyard during king tides / heavy rain; 2024 storm-surge debris in inlets.	Repair — included in FY27 needs (not a lifecycle component)	$189,000	Yes
TRM-01	Tarpon Ridge MS	HVAC	BAS overridden: 41% of AHUs in hand, after-hours runtime 62 hrs/wk beyond occupancy.	Retro-commissioning, including repair of the failed air-handler economizer dampers. Repair — included in FY27 needs (not a lifecycle component)	$124,000	Yes
MBH-02	Mangrove Bay HS	Plumbing	Original 1958 galvanized piping in A-building.	Full component renewal	$2,731,000	Yes
MBH-04	Mangrove Bay HS	Roofing	Campus roofs (2005) beyond service life with active leaks; worst at the gymnasium, where 2024 storm damage was only partially repaired.	Full component renewal	$3,604,000	Yes

ID	SITE	SYSTEM	FINDING	SCOPE	EST. PROJECT COST	IN 10-YR FORECAST
MBH-05	Mangrove Bay HS	Roofing	Gymnasium roof deck corroded at drains (approx. 15% of 18,000 sf gym roof).	Repair — included in FY27 needs (not a lifecycle component)	$70,000	Yes
WSH-01	Westshore HS	Exterior Enclosure	Stucco cracking and sealant failure on west/south elevations; salt-air spalling at balconies.	Repair — included in FY27 needs (not a lifecycle component)	$897,000	Yes
WSH-02	Westshore HS	Sitework	Site straddles Zones VE and AE; the ground-floor main electrical room is in the Zone VE wing. 2024 surge reached it.	Repair — included in FY27 needs (not a lifecycle component)	$2,080,000	Yes
DOC-01	Operations & Transportation Ctr	Roofing	Metal roof fasteners corroded; panel uplift at eaves.	Full component renewal	$1,638,000	Yes
Total					**$19.7M**	

11.3 Phasing

FIGURE 19

Phasing, FY27–FY36



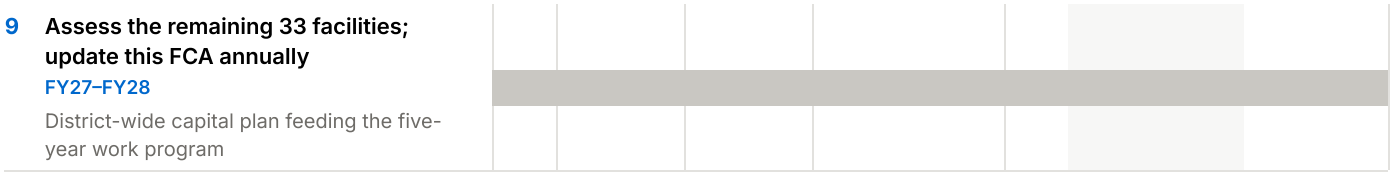
Phase 1 — Stabilize & Retrofit FY27–FY28	All Priority 1 items; Tarpon Ridge RCx; HVAC and controls renewals delivered as a performance contract, with monitoring at Coral Key and Mangrove Bay HS; roofs at end of life at Mangrove Bay HS (campus), Westshore HS, and the Operations Center ($10.2M; Coral Key held pending study); Bayside windows and Westshore envelope repairs; Westshore electrical relocation out of Zone VE; Coral Key replace-vs-renovate study	**$88.9M** incl. $6.2M of Coral Key scope on hold
Phase 2 — Complete the Envelope FY29–FY31	Remaining roofs and impact windows (Mangrove Bay HS, Operations Center); electrical renewals; Tarpon Ridge air-handler replacement	**$67.0M**
Phase 3 — Sustain FY32–FY36	Interiors, plumbing, site; second-cycle HVAC; extend monitoring to more sites as costs fall	**$53.2M**

11.4 Next Steps

FIGURE 20

Next steps, from board briefing to construction





Week 0 is the board briefing. Shaded band: summer 2027 construction window. Dashed bar: 40–50-week lead time for electrical equipment after each scope is approved.

SECTION 12

Site Reports

Each site report follows the same template: profile, condition summary, major-scope quantities, system inventory, utility benchmarking, deficiencies, and measures. Coral Key Elementary is shown in full.

SITE REPORT 12.1

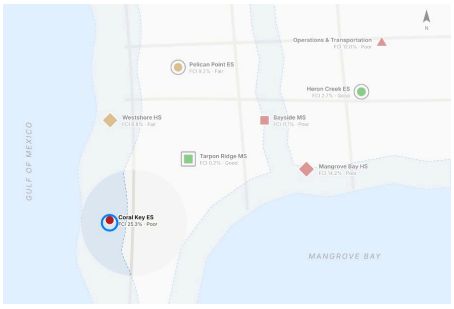
Coral Key Elementary



Coral Key Elementary, exterior. AI-generated illustration for the fictional sample FCA.



Coral Key Elementary, aerial context. AI-generated illustration for the fictional sample FCA.



Locator: Coral Key Elementary (circled) within the district. Schematic map of a fictional district.

PROFILE

Address	1410 Coral Key Blvd, Mangrove Bay, FL
Grades / Enrollment	PK-5 / 612
Year built / Last major renovation	1962 / 1998 (partial: HVAC, ceilings, fire sprinklers)
Gross area / Stories	68,400 sf / 1
Site area	11.2 acres
Flood zone / Coastal exposure	AE / Yes (< 1 mile)
EHPA shelter	No
Current replacement value	$29.1M ($425/sf)

CONDITION SUMMARY

Current needs	**$7.3M**
FCI	**25.3% — Poor** (highest in portfolio; 2.7× portfolio average)
10-year renewal need	**$14.7M** (51% of CRV)
EUI (FY25)	**58.9 kBtu/sf** (second-highest of 8)
Quadrant	**Renew & Retrofit**

Assessor's summary. Coral Key is a 1962 single-story, finger-plan campus with exterior covered walkways, typical of its era in coastal Florida. It has been kept in service through partial renovations, but much of the original building fabric is now the limiting factor:

- **Failed or poor:** the single-pane steel windows leak at the sills, the galvanized domestic water piping delivers low pressure and discolored water, the 2003 roof has failed in multiple locations, and three of fourteen rooftop units have lost their condenser coils to salt air.
- **Classroom conditions:** rooms on the 100-wing were above 78°F and 65% relative humidity during the assessment, a mold and instructional-quality risk as well as a comfort problem.

- **Old but still working:** several original elements, including interior doors, sanitary piping, and storm drainage, are well beyond their expected life but still functioning. They are rated Fair and flagged for evaluation, not for replacement.

Recommendation: stabilize, then decide. Coral Key's FCI, its age, its Flood Zone AE location, and the educational limits of a 1962 finger plan together justify a replace-vs-renovate study. The 10-year renewal need ($14.7M, 51% of CRV) is one input to that study, not its conclusion. We recommend:

1. **Proceed now with $1.2M:** the Priority 1 items ($0.4M: replace the three failed rooftop units with coastal-coated units, add dehumidification for the 100-wing, correct the accessible route, and repair the fire sprinklers); domestic water piping, CKE-05 ($0.6M, water quality); and the emergency generator ($0.2M, life safety).
2. **Commission a replace-vs-renovate study in FY27** that compares both paths over the same 30-year horizon:
   - capital cost, including demolition and temporary student accommodations for the replacement option
   - lifecycle renewal and maintenance
   - energy and operating costs
   - flood-mitigation requirements under each path
   - educational adequacy and enrollment projections
   - residual value at the end of the period
3. **Hold $6.2M of renewal scope until the study is complete:** roof ($2.37M), windows ($1.15M), the rest of the rooftop-unit fleet ($0.69M), the rest of air handling and DOAS ($1.02M), ceilings ($0.55M), and communications and security ($0.40M). Fund an interim roof-restoration allowance of about $0.3M (targeted repairs and coating, 73,900 sf at $3.00/sf plus coastal premium and soft costs) to keep the building dry for 24–36 months. The allowance is excluded from the needs totals; it is the cost of deferring the decision.

MAJOR-SCOPE QUANTITIES

SCOPE	QUANTITY	UNIT COST (INCL. COASTAL)	CONSTRUCTION	PROJECT COST (×1.30)	NOTES
Roof coverings & flashing	73,900 roof sf	$24.64	$1,821,000	$2,367,000	Footprint × 1.08 for overhangs and covered walks; tear-off, tapered ISO to R-30 ci, modified bitumen; deck repair allowance excluded
Exterior windows & storefront	6,840 glazing sf	$128.80	$881,000	$1,145,000	~10% of GSF; impact-rated aluminum, installed with removal
Packaged DX / rooftop units	155 tons	$4,368.00	$677,000	$880,000	Classroom wings (~75% of area) at ~330 sf/ton; ~14 units averaging 11 tons; includes curb adapters, crane, electrical reconnection, controls integration. Excludes ductwork and outdoor air (see next row)
Air handling, DOAS & ductwork	14,300 OA cfm (DOAS)	$63.13	$903,000	$1,174,000	Outdoor air = 10 cfm/student + 0.12 cfm/sf (ASHRAE 62.1 basis); DOAS at $30/cfm plus the 1985 air handlers, ductwork and exhaust at $5.50/sf. Excludes packaged units
Emergency power (generator & ATS)	125 kW	$1,064.00	$133,000	$173,000	Life-safety load ~2 W/sf; diesel generator with ATS
Electrical service & distribution	1,200 amps	$627.20	$753,000	$979,000	Service sized at ~13 W/sf, 480V; switchboard, panelboards, feeders

HVAC systems and scope boundaries. Packaged rooftop units serve classroom wings (~75% of area); original 1985 air handlers with 1998 VAV boxes and gas-boiler hot water serve cafeteria, media center and admin (~25%); DOAS units supply classroom outdoor air. The cost categories follow those zones: packaged rooftop units (D3032) cover the classroom-wing equipment only; D3040 carries the DOAS units, the 1985 air handlers, ductwork, and exhaust; terminal units (D3050) and the gas boiler (D3020) are priced for the ~25% of the building they serve. No equipment is counted in more than one category, and the Priority 1 scope is split the same way (CKE-01 rooftop units, CKE-02 DOAS).

SYSTEM INVENTORY

UNIFORMAT	COMPONENT	INSTALLED	EUL	RUL	OBSERVED CONDITION	LIFECYCLE STATUS	BUDGET YEAR	PROJECT COST
B2010	Exterior walls - sealants, stucco & coatings	2004	13	2	Fair	Beyond EUL - evaluate	FY28	$896,000
B2020	Exterior windows & storefront	1962	35	-29	Poor	Beyond EUL - evaluate	FY27	$1,145,000
B2030	Exterior doors & hardware	1998	22	3	Fair	Beyond EUL - evaluate	FY29	$250,000
B3010	Roof coverings & flashing	2003	20	-3	Poor	Beyond EUL - evaluate	FY27	$2,367,000
C1020	Interior doors & hardware	1962	40	8	Fair	Beyond EUL - evaluate	FY34	$299,000
C3010	Wall finishes	2012	12	2	Fair	Beyond EUL - evaluate	FY28	$348,000
C3020	Floor finishes	2005	15	4	Fair	Beyond EUL - evaluate	FY30	$748,000
C3030	Ceiling finishes	1998	25	-3	Poor	Beyond EUL - evaluate	FY27	$547,000
D2010	Plumbing fixtures	1985	35	6	Fair	Beyond EUL - evaluate	FY32	$547,000
D2020	Domestic water distribution	1962	40	-24	Poor	Beyond EUL - evaluate	FY27	$598,000
D2023	Domestic water heaters	2008	15	5	Fair	Beyond EUL - evaluate	FY31	$120,000
D2030	Sanitary waste & vent	1962	50	6	Fair	Beyond EUL - evaluate	FY32	$449,000
D3020	Heat generation (gas boilers, hot water)	1998	25	4	Fair	Beyond EUL - evaluate	FY30	$74,000
D3032	Packaged DX / rooftop units	2010	12	-4	Poor	Beyond EUL - evaluate	FY27	$880,000
D3040	Air handling, DOAS & ductwork	1985	30	-11	Poor	Beyond EUL - evaluate	FY27	$1,174,000
D3050	Terminal units (VAV / FCU)	1998	25	7	Fair	Beyond EUL - evaluate	FY33	$125,000
D3060	Building automation & controls	2006	15	5	Fair	Beyond EUL - evaluate	FY31	$449,000
D4010	Fire sprinkler systems	1999	50	23	Good	Within life	2049 (beyond plan)	$449,000
D5010	Electrical service & distribution	1985	45	3	Fair	Within 5 yrs of EUL	FY29	$979,000
D5020	Lighting & branch wiring	2014	20	8	Good	Within life	FY34	$846,000
D5037	Fire alarm & mass notification	2009	15	4	Fair	Beyond EUL - evaluate	FY30	$348,000
D5038	Communications & security (access, CCTV)	2015	12	1	Fair	Within 5 yrs of EUL	FY27	$398,000
D5092	Emergency power (generator & ATS)	1999	22	-5	Poor	Beyond EUL - evaluate	FY27	$173,000
E1093	Food service equipment	2005	15	4	Fair	Beyond EUL - evaluate	FY30	$299,000
G2020	Parking lots & drives	2001	20	2	Fair	Beyond EUL - evaluate	FY28	$300,000
G2030	Pedestrian paving & accessible routes	1999	30	3	Fair	Within 5 yrs of EUL	FY29	$148,000
G3030	Storm drainage	1962	40	5	Fair	Beyond EUL - evaluate	FY31	$148,000

Observed condition is the assessor's field rating. Lifecycle status is age-based. An item beyond its expected useful life (EUL) but still functioning is rated Fair and flagged for evaluation; it is not budgeted for replacement on age alone.

Condition by system group

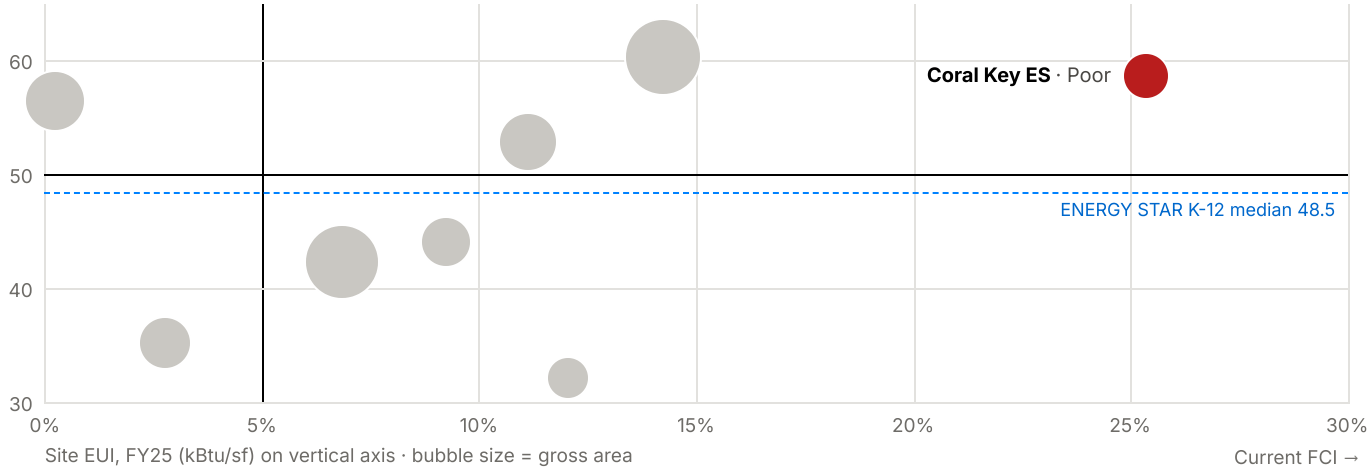
SYSTEM	WORST OBSERVED CONDITION	DUE FY27	10-YEAR TOTAL
Exterior Enclosure	Poor	$1,145,000	$2,291,000
Roofing	Poor	$2,367,000	$2,367,000
Interiors	Poor	$547,000	$1,942,000
Plumbing	Poor	$598,000	$1,714,000
HVAC	Poor	$2,054,000	$2,702,000
Fire Protection	Good	$16,000	$16,000
Electrical	Poor	$571,000	$2,744,000
Equipment	Fair	$0	$299,000
Sitework	Fair	$0	$596,000
Accessibility	— (repair item)	$49,000	$49,000
Total		**$7,347,000**	**$14,720,000**

UTILITY BENCHMARKING

METRIC	FY22	FY23	FY24	FY25	TREND
Electricity (kWh)	1,145,700	1,128,800	1,097,600	1,098,100	
Natural gas (therms)	2,940	2,900	2,820	2,820	
Water (kgal)	2,402	2,272	2,271	2,304	
Energy cost	$123,920	$145,130	$134,790	$133,670	
Total utility cost	$148,900	$169,670	$160,450	$160,850	
EUI (kBtu/sf)	61.4	60.5	58.9	58.9	
Energy cost / sf	$1.81	$2.12	$1.97	$1.95	

FIGURE 21

Coral Key Elementary within the portfolio EUI vs. FCI distribution



Coral Key's EUI of 58.9 kBtu/sf is the second-highest of 8 in the district, and its utility cost per square foot ($2.35) is the highest. Usage has drifted down slightly since FY22 as failed rooftop units were replaced piecemeal. Single-pane glazing and a poorly insulated roof add solar and moisture load, but the modeled energy value of fixing them is small. The case for the roof and windows is water intrusion, hurricane resilience, and indoor humidity, not energy.

DEFICIENCIES



AI-generated illustration for the fictional sample FCA.

CKE-01 PRIORITY 1 HVAC

Finding	RTU-4, -7 and -11 condenser coils fully corroded; units tripping on high head pressure.
Recommendation	Replace three RTUs with coastal-coated units.
Cost	$189,000 for a partial scope: 3 of 14 RTUs (approx. 33 of 155 tons). The balance of the packaged DX / rooftop units is in the lifecycle forecast
Driver	Comfort, instructional disruption · Linked to FIM



AI-generated illustration for the fictional sample FCA.

CKE-02 PRIORITY 1 HVAC

Finding	Classrooms 104-118 above 78°F / 65% RH during walk-through; the 100-wing (8 classrooms, ~7,200 sf) has no dedicated outdoor-air dehumidification.
Recommendation	Add a ~3,000 cfm DOAS (ASHRAE 62.1 classroom rate) for the 100-wing: unit with hot-gas reheat, roof curb and structural support, ducted outdoor air to existing classroom supply, electrical, and BAS integration. Construction about $116K: DOAS at $30/cfm with coastal premium ($101K), plus about $15K for curb, structure and connections.
Cost	$151,000 for a partial scope: 100-wing DOAS only, ~3,000 cfm. The balance of the air handling, DOAS & ductwork is in the lifecycle forecast
Driver	Moisture/IAQ · Linked to FIM



AI-generated illustration for the fictional sample FCA.

CKE-07 PRIORITY 1 Accessibility

Finding	Accessible route from bus loop to main entry has 4.1% cross-slope and a 1.5" vertical change at sidewalk joint.
Recommendation	Regrade and replace sidewalk segment; detectable warnings at curb ramp.
Cost	$49,000 · Repair — included in FY27 needs (not a lifecycle component)
Driver	ADA / 2012 Standards



AI-generated illustration for the fictional sample FCA.

CKE-08 PRIORITY 1 Fire Protection

Finding	Sprinkler heads in kitchen hood area painted over; 5-year internal pipe inspection overdue.
Recommendation	Replace painted heads; perform NFPA 25 internal inspection.
Cost	$16,000 · Repair — included in FY27 needs (not a lifecycle component)
Driver	Life safety (SREF)



AI-generated illustration for the fictional sample FCA.

CKE-04 PRIORITY 2 Roofing

Finding	Modified bitumen roof (2003) at end of life; ponding over cafeteria, 14 patched leaks on work-order history.
Recommendation	Full tear-off, tapered insulation to R-30 ci, reroof.
Cost	$2,367,000: full component renewal, carried in the lifecycle forecast (FY27)
Driver	Water intrusion, energy · Linked to FIM



AI-generated illustration for the fictional sample FCA.

CKE-03 PRIORITY 2 Exterior Enclosure

Finding	Original 1962 single-pane steel windows in 100/200 wings; non-impact glazing, failed glazing seals and water intrusion at sills.
Recommendation	Replace with impact-rated aluminum windows designed to FBC 9th Ed. (2026) wind-borne debris requirements.
Cost	$859,000 for a partial scope: 100/200 wings, approx. 75% of glazing. The balance of the exterior windows & storefront is in the lifecycle forecast
Driver	Hurricane resilience, water intrusion · Linked to FIM



AI-generated illustration for the fictional sample FCA.

CKE-05 PRIORITY 2 Plumbing

Finding	Original galvanized domestic water piping; low pressure and rust-colored water at far fixtures.
Recommendation	Replace domestic water distribution with copper/PEX.
Cost	$598,000: full component renewal, carried in the lifecycle forecast (FY27)
Driver	Water quality, reliability



AI-generated illustration for the fictional sample FCA.

CKE-06 PRIORITY 3 Electrical

Finding	1985 main switchboard; replacement parts no longer manufactured, no arc-flash labeling.
Recommendation	Replace main switchboard and panelboards; arc-flash study.
Cost	$979,000: full component renewal, carried in the lifecycle forecast (FY29)
Driver	Reliability, NFPA 70E



AI-generated illustration for the fictional sample FCA.

CKE-09 PRIORITY 4 Electrical

Finding	2014 fluorescent-to-LED retrofit lamps (T8 LED tubes) on original ballasts/fixtures; no occupancy controls.
Recommendation	Full LED fixture replacement with networked controls at next renewal or as part of the replace-vs-renovate outcome.
Cost	$846,000: full component renewal, carried in the lifecycle forecast (FY34)
Driver	Energy

APPLICABLE FACILITY IMPROVEMENT MEASURES

FIM	MEASURE	BASIS	INSTALLED COST	NET SAVINGS / YR	PAYBACK ON EFFICIENCY COST	NOTE
F1	HVAC replacement - coastal-coated rooftop units (IEER 14.5)	33 tons; effective EER 9.5 to IEER 14.5; EC fan motors (3 failed units only, pending study)	$189,000	$3,195	7.7 yrs	Stabilization scope only; full fleet decision follows the study
F4	Retro-commissioning (schedules, setpoints, resets)	8% of HVAC kWh: schedule, setpoint and reset corrections	$53,000	$4,832	11.0 yrs	Proceed; low cost, informs study
F8	Water conservation (fixtures, smart irrigation)	18% of water & sewer cost (fixtures, irrigation controllers)	$53,000	$4,892	10.8 yrs	Proceed; savings independent of the building's future
F10	Monitoring-based commissioning (continuous analytics)	5% of remaining HVAC kWh (fault detection); persistence of controls savings not credited	$13,000	$1,068	12.2 yrs	Proceed; trend data informs the study

Roof insulation (F6), window replacement (F7), and full rooftop-fleet replacement are held pending the replace-vs-renovate study and are not counted in the measures above. LED fixture replacement is not recommended as a standalone measure: the 2014 LED tube retrofit already captured most of the savings, and new fixtures belong in whichever path the study selects.

12.2–12.8 Remaining Site Reports

Abbreviated in this sample. Data for each site is in the companion workbook; full reports follow the 11.1 template.

SECTION	SITE	FCI	QUADRANT	HEADLINE
12.2	Pelican Point Elementary	9.2%	Capital Renewal	EHPA shelter with failing coastal rooftop units and an undersized generator
12.3	Heron Creek Elementary	2.7%	Maintain	Good today; 2006 chillers and systems approach their first renewal cycle
12.4	Bayside Middle School	11.1%	Renew & Retrofit	Single chiller with no redundancy; 1971 non-impact windows
12.5	Tarpon Ridge Middle School	0.2%	Optimize	Best condition, high EUI: retro-commissioning is the top return in the district
12.6	Mangrove Bay High School	14.2%	Renew & Retrofit	Largest backlog: central plant, 1958 piping, elevators, roof
12.7	Westshore High School	6.8%	Capital Renewal	Fair FCI understates flood risk: electrical relocation/elevation out of Zone VE
12.8	District Operations & Transportation Center	12.0%	Capital Renewal	Support facility with concentrated renewals ahead

SECTION 13

Conclusion

Mangrove Bay's buildings are not in crisis as a portfolio. At 9.5% FCI, they rate Fair. But the average hides a concentrated backlog, a maintenance budget well below the state norm, and a renewal cycle that, after FY27, will require about $17.3M a year at just eight of 41 facilities. Without a deliberate plan, the gap widens every year and the district's newest schools become the next backlog.

The path forward is practical:

- **Fix what has failed.** Fund the Priority 1 items now ($7.2M).
- **Run what works properly.** Retro-commission Tarpon Ridge, and commission the HVAC systems the district is about to buy so they perform as designed.
- **Renew the rest as one program.** Deliver HVAC and controls renewals as an integrated, guaranteed program; fund the envelope and resilience work from capital outlay and hazard-mitigation grants.
- **Decide Coral Key on evidence.** Hold $6.2M of renewal scope there and let the replace-vs-renovate study settle its future.

Energy savings help, but they will not fund the backlog. Clear priorities, honest numbers, and a sustained funding plan will.

APPENDIX A

Methodology

1. Data collection. We reviewed district-provided records: FISH (Florida Inventory of School Houses) data, as-built drawings, CMMS work-order history (36 months), SREF annual safety inspection reports, prior capital plans, and four fiscal years of utility bills.

2. Field assessment. A team of an architect, a mechanical engineer, and an electrical engineer, supported by Upchurch HVAC and controls technicians, walked every building. We observed roofs, mechanical rooms, and electrical rooms, plus at least 25% of classrooms at each site. Equipment nameplates were recorded, and building automation trends were reviewed where available.

3. Component inventory. Each building was broken into systems using ASTM UNIFORMAT II (Level 3). For each component we recorded install year, expected useful life (EUL, adjusted for coastal exposure), observed condition, and remaining useful life (RUL). Components beyond EUL but functioning were given an observation-based life extension of up to 10 years and flagged for annual evaluation.

4. Quantities and costs. (Unit-cost inclusions and the replacement-value derivation are in Appendix C.) Major scopes were priced from quantities: roof area, glazing area, cooling tonnage, outdoor-air capacity, generator kW, and electrical service amperage. Each used Florida-adjusted 2026 unit costs. Minor systems were priced per gross square foot. A 1.30 multiplier converts construction cost to project cost (design, general conditions, contingency), the same basis used for CRV.

5. Deficiency identification. Conditions requiring action beyond normal lifecycle renewal were documented with photographs, a priority, a recommended corrective action, a defined scope (full, partial, or repair), and an estimated cost.

6. Energy analysis. Utility data was normalized to EUI and cost per square foot, then benchmarked within the portfolio, against the ENERGY STAR national K-12 median, and against FL DOE district energy-cost data. Savings potential was modeled by end use with measures applied in sequence and recurring costs deducted (Section 10).

7. Cost–benefit workshops. The stakeholder groups listed in Section 3 weighted ten cost and benefit criteria and scored every facility improvement measure 1–10 on each. Weighted scores position measures on a cost–benefit matrix; need-based measures proceed regardless of score.

8. Prioritization. Needs were ranked by consequence of failure, then phased to align with Florida's five-year District Facilities Work Program cycle.

APPENDIX B

Rating Definitions

Observed condition (assessor rating)

RATING	DEFINITION
Good	Performing as designed; normal wear
Fair	Aging or beyond EUL but functional; increasing maintenance; evaluate annually
Poor	Degraded performance or reliability; failure likely within two years
Failed	Not functioning, unsafe, or in active failure

Lifecycle status (age-based)

STATUS	DEFINITION
Within life	More than 5 years of expected life remaining
Within 5 yrs of EUL	Plan renewal funding
Beyond EUL — evaluate	Past expected life; replace only when observed condition warrants

FCI rating: Good ≤ 5%, Fair 5–10%, Poor 10–30%, Critical > 30%.

10-year renewal need / CRV: cumulative FY27–FY36 spending ÷ CRV. A funding-scale measure, not a condition rating.

APPENDIX C

Cost Basis & Assumptions

ITEM	VALUE
Cost basis	2026 dollars, no escalation; planning-level (±25%)
CRV definition	Total project cost to replace in kind to current code: building and immediate site construction, design, general conditions, contingency. Excludes land, FF&E, demolition, temporary housing
CRV ($/sf)	Elementary $425 · Middle $445 · High $510 · Support $320. Derivation in the table below
Project-cost multiplier	1.30× construction (same basis as CRV): design and engineering 8%, construction and owner contingency 10%, occupied-school phasing and general conditions 7%, permits, testing, special inspections and commissioning 3%, owner administration 2%
Coastal premium	+12% on sites < 1 mile from coast
Flood screening elevation	Critical equipment screened against BFE + 1 ft. Final design elevations follow the Florida Building Code (ASCE 24 flood design class) and the local floodplain ordinance, which may require more freeboard
Roofing	$22.00/roof sf (metal re-roof at support facility $24.00). Includes tear-off, new insulation to current energy code, membrane, flashing and wind-uplift attachment. Deck repair excluded except where a deficiency carries it (MBH-05)
Impact glazing	$115/glazing sf. Impact-rated windows and storefront with frames, installation and perimeter sealants
Packaged rooftop HVAC	$3,900/ton. Coastal-coated unit, curb adapter, crane, electrical and controls reconnection. Ductwork excluded
Chilled-water plant	$2,900/ton. Complete plant: chillers, cooling towers (water-cooled sites), chilled- and condenser-water pumps with VFDs, plant piping and controls. Distribution piping and air handlers excluded (air handlers carried separately)
DOAS	$30/cfm outdoor air, plus air distribution $5.50/sf (DX sites) or $11.00/sf (chiller sites, incl. AHUs)
Generator with ATS	$950/kW. Generator, automatic transfer switch, wind-rated enclosure, fuel system and pad
Electrical service & distribution	$560/amp of service capacity. Service switchboard, main distribution and feeders to panelboards. Branch panels and wiring carried under lighting and branch wiring ($8.50/sf)
Electricity (FY25 blended)	$0.118/kWh; efficiency savings valued at $0.110/kWh; PV output at $0.085/kWh
Natural gas / water & sewer (FY25)	$1.45/therm / $11.80/kgal
Solar PV	$2,750/kWdc installed; 1,450 kWh/kWdc-yr; $20/kWdc-yr O&M; storage not included
LED fixtures & networked controls	$2.40/sf (fluorescent sites only)
Monitoring-based commissioning	$0.15/sf implementation; $0.025/sf-yr analytics
M&V	3% of guaranteed savings (performance-contract measures)
Financing basis	20 years (F.S. 1013.23 maximum), 5%
Benchmarks	ENERGY STAR US median site EUI, K-12: 48.5 kBtu/sf; FL DOE 2024–25 energy cost per GSF: $1.35 statewide; Charlotte $1.41, Pinellas $1.53, Collier $1.61 (FL DOE, *Florida School District Annual Energy Cost Information, 2024–2025*, "All Energy" cost per gross square foot: https://cdn.fldoe.org/file/5599/AnnualEnergyReport2425.pdf)

How the replacement values were derived. FL DOE's *Cost of Construction* report for calendar year 2024 lists 23 new elementary projects at an average total plant cost of $455/GSF (simple average of projects) or $409/GSF (weighted by floor area). Total plant cost includes design fees, construction, furniture and equipment, and site improvements.

STEP	ELEMENTARY $/GSF
FL DOE 2024 total plant cost, elementary (weighted – simple average)	$409 – $455
Remove furniture and equipment (4.9% of total plant cost), which CRV excludes	$389 – $433
Escalate two years at 4% to 2026	$421 – $468
Adopted CRV, elementary	**$425** (conservative end of the range)

FL DOE's 2024 middle- and high-school averages ($390 and $402/GSF, ten projects each) are dominated by additions and small projects, so they understate full-building replacement. Middle, high and support values are set as ratios to the elementary value: **middle 1.05×** ($445), **high 1.20×** ($510; athletics, auditorium, career-and-technical labs and central plants), and **support 0.75×** ($320; pre-engineered and shop construction). Source: FL DOE, Office of Educational Facilities, Cost of Construction, Public School Districts, calendar year 2024 (https://www.fldoe.org/core/fileparse.php/7742/urlt/24CostConst-PS.pdf).

APPENDIX D

Glossary

- **BFE:** Base Flood Elevation (FEMA)
- **CRV:** Current Replacement Value
- **DOAS:** Dedicated Outdoor Air System, used for humidity control
- **EHPA:** Enhanced Hurricane Protection Area, a designated shelter space in Florida public schools
- **EUI:** Energy Use Intensity, annual site energy per square foot (kBtu/sf/yr)
- **EUL / RUL:** Expected Useful Life / Remaining Useful Life
- **FCI:** Facility Condition Index, current needs ÷ CRV
- **FIM:** Facility Improvement Measure
- **FISH:** Florida Inventory of School Houses
- **IEER:** Integrated Energy Efficiency Ratio, the part-load efficiency rating for packaged cooling
- **kWdc:** solar array capacity, kilowatts direct current
- **M&V:** Measurement and Verification of guaranteed savings
- **PECO:** Public Education Capital Outlay
- **RCx / MBCx:** Retro-commissioning / Monitoring-based commissioning
- **SREF:** State Requirements for Educational Facilities
- **UNIFORMAT II:** ASTM E1557 classification of building elements

APPENDIX E

Facility Improvement Measure Descriptions

1.1 Packaged rooftop unit replacement (coastal-coated)

Mechanical · CKE, PPE, DOC · Phase 1 **Need-based**

Replace failed and end-of-life packaged rooftop units with high-efficiency (IEER ≥ 14.5) units carrying factory coastal coil coatings. Coral Key is limited to the three failed units pending the replace-vs-renovate study.

1.2 Chiller plant replacement & optimization

Mechanical · HCE, BMS, MBH, WSH · Phase 1 **Need-based**

Replace chillers, towers, and pumps at end of life with a high-efficiency plant (variable-speed compressors and pumping, optimized chilled-water reset). Bayside and Mangrove Bay HS are failed or failing; Heron Creek's air-cooled plant is replaced like-for-like.

1.3 Cooling tower refurbishment

Mechanical · TRM · Phase 2

Rebuild cooling tower basins, fill, and drift eliminators; add VFD fan control and water-treatment upgrades at water-cooled plants not scheduled for full replacement.

1.4 Dedicated outdoor-air (DOAS) dehumidification

Mechanical · CKE, BMS, MBH, TRM · Phase 1 **Need-based**

Add dedicated outdoor-air units to dehumidify ventilation air separately from space cooling, controlling classroom humidity below 60% RH. Coral Key's 100-wing is the priority.

1.5 Air-handler replacement with VAV conversion

Mechanical · TRM, BMS, MBH · Phase 2

Replace original constant-volume air handlers with variable-air-volume units, DOAS integration, and working economizers. Tarpon Ridge's 1998 units are the first candidates.

1.6 Heat-pump water heaters

Mechanical · CKE, PPE, DOC · Phase 3

Replace gas or electric-resistance storage water heaters at end of life with heat-pump water heaters, which also provide incidental cooling and dehumidification in mechanical rooms.

2.1 Building automation system upgrade

Controls & Energy Mgmt · PPE, HCE, BMS, MBH, DOC · Phase 1

Replace obsolete building automation systems with open-protocol controls, rebuilt schedules, setpoint and reset strategies, and commissioning. Integrates with district-wide monitoring.

2.2 Retro-commissioning

Controls & Energy Mgmt · CKE, TRM, WSH · Phase 1 **Need-based**

Test, adjust, and correct existing systems without major equipment replacement: schedules, overrides, setpoints, economizers, and sequences. Tarpon Ridge has documented overrides and is the first site; it proceeds now as a standalone project, outside the performance contract.

2.3 Monitoring-based commissioning (continuous analytics)

Controls & Energy Mgmt · CKE, MBH · Phase 1

Continuous analytics on BAS and meter data to detect faults and keep savings from drifting. Recommended at Coral Key and Mangrove Bay HS, where direct savings repay the cost within 15 years; elsewhere it is an operations tool to be justified on those terms.

2.4 Kitchen hood demand-controlled ventilation

Controls & Energy Mgmt · BMS, MBH, WSH · Phase 2

Variable-speed kitchen exhaust and makeup air driven by cooking activity at schools with full production kitchens.

2.5 Utility bill management & tariff review

Controls & Energy Mgmt · All sites · Phase 1

Audit utility bills and tariffs monthly for errors, rate-class optimization, and demand charges; feed results to the energy manager.

2.6 Student & staff energy teams

Controls & Energy Mgmt · All sites · Phase Optional

Student and staff energy teams at each school, with dashboards and simple behavior campaigns; low cost, modest savings, strong educational value.

3.1 LED fixtures with networked controls (fluorescent sites)

Lighting · HCE, BMS, MBH · Phase 2

Replace remaining fluorescent fixtures with LED fixtures and networked occupancy and daylight controls. Limited to the three sites that have not already retrofitted.

3.2 Exterior & parking-lot LED

Lighting · All sites · Phase 2

Convert exterior building and parking-lot lighting to LED with photocell and schedule control; improves after-hours safety.

3.3 Athletic field LED

Lighting · MBH, WSH · Phase Optional

Convert athletic field lighting to LED with instant-on control; reduces demand charges and spill light for neighbors.

4.1 Switchgear & distribution modernization

Electrical & Power · DOC (CKE pending study) · Phase 2

Replace obsolete switchboards and panelboards where parts are no longer available; includes arc-flash study and labeling. The Operations Center first; Coral Key only if the study favors renovation. Order early given current lead times.

4.2 EHPA shelter generator upsizing

Electrical & Power · PPE · Phase 1 **Need-based**

Upsize the shelter generator and transfer switch at Pelican Point so the EHPA shelter area can run full HVAC during an outage.

4.3 Rooftop solar PV (shelter sites)

Electrical & Power · PPE, HCE, TRM · Phase Optional

Rooftop solar arrays at the three shelter schools. A weak stand-alone financial case at current tariffs; considered mainly when paired with storage.

4.4 Battery storage & islanding for shelters

Electrical & Power · PPE, HCE, TRM · Phase Optional

Battery storage with islanding controls so shelter-site solar can support critical loads during a grid outage. Federal credits for storage continue; pursue with resilience grants.

4.5 Electric school bus charging

Electrical & Power · DOC · Phase Optional

Charging infrastructure for electric school buses at the Operations Center. Defer unless grant funding or a fleet-replacement decision changes the economics.

5.1 Roof replacement with insulation upgrade

Envelope · PPE, HCE, BMS, MBH, WSH, DOC · Phase 1–2

Need-based

Full roof replacement with tapered insulation to R-30 continuous and FBC 9th Edition wind-uplift attachment. Phase 1 covers roofs at end of life (Mangrove Bay HS campus and the Operations Center, both need-based, and Westshore); Phase 2 covers the remaining sites. Coral Key is held pending the study.

5.2 Impact-rated window replacement

Envelope · BMS, MBH, DOC · Phase 1–2 **Need-based**

Replace non-impact single-pane windows with impact-rated, low-e insulated glazing; removes the need for stored shutters. Bayside (need-based) in Phase 1; Mangrove Bay HS and the Operations Center in Phase 2.

5.3 Exterior wall restoration & sealants

Envelope · WSH · Phase 1 **Need-based**

Restore stucco, sealants, and coatings on deteriorated elevations to stop water intrusion; Westshore's west and south elevations first.

6.1 Fixture retrofits & smart irrigation

Water & Plumbing · All sites · Phase 1

Low-flow fixtures and smart irrigation controllers at all sites; quick payback and low disruption.

6.2 Domestic water piping replacement

Water & Plumbing · CKE, MBH · Phase 1 **Need-based**

Replace original galvanized domestic water piping with copper or PEX at Coral Key and Mangrove Bay HS (A-building).

6.3 Water submetering & leak detection

Water & Plumbing · All sites · Phase 3

Submeter irrigation, kitchens, and large water users, with leak-detection alarms.

7.1 Elevate or relocate critical electrical

Resilience & Site · WSH, MBH · Phase 1 **Need-based**

Relocate or elevate main electrical service above the design flood elevation (screened here at base flood elevation plus 1 ft). Westshore is need-based; its electrical room is in Zone VE, where dry floodproofing is not permitted, so relocation is the path. Mangrove Bay HS (Zone AE) critical equipment should be raised during scheduled renewals.

7.2 Storm drainage & backflow prevention

Resilience & Site · BMS, CKE · Phase 1 **Need-based**

Clean and video-inspect storm lines; add backflow preventers and inlet protection where tidal and storm backup occurs (Bayside courtyard).

7.3 Wind-rated rooftop equipment anchorage

Resilience & Site · All coastal sites · Phase 2

Re-anchor rooftop equipment to current wind-load requirements at coastal sites, usually alongside unit replacement or reroofing.

8.1 Access control & secure entry vestibules

Safety & Security · All sites · Phase 2

Electronic access control and secure entry vestibules at every school, integrated with visitor management.

8.2 Fire alarm & mass notification modernization

Safety & Security · WSH, DOC, CKE · Phase 1

Replace end-of-life fire alarm panels with addressable systems and integrated mass notification.

8.3 Camera system upgrade

Safety & Security · All sites · Phase 3

Upgrade analog and early IP cameras to current IP systems with adequate retention and coverage of entries and parking.

APPENDIX F

Cost–Benefit Workshop Scorecard

Workshop scores behind the cost–benefit matrix in Section 10.2. Cost criteria are shaded orange and benefit criteria blue; weights are shown in each column heading.

FIGURE 22

Workshop scorecard: cost criteria

FIM	Measure	Capital (40%)†	Staff time (20%)	Disruption (20%)	O&M burden (10%)	Risk (10%)	Cost score
1.1	Packaged rooftop unit replacement (coastal-coated)	5	4	4	2	3	**4.10**
1.2	Chiller plant replacement & optimization	9	6	5	3	6	**6.70**
1.3	Cooling tower refurbishment	2	3	2	1	2	**2.10**
1.4	Dedicated outdoor-air (DOAS) dehumidification	7	5	5	4	5	**5.70**
1.5	Air-handler replacement with VAV conversion	9	5	6	2	5	**6.50**
1.6	Heat-pump water heaters	2	2	1	2	2	**1.80**
2.1	Building automation system upgrade	6	7	2	4	3	**4.90**
2.2	Retro-commissioning	2	5	1	1	1	**2.20**
2.3	Monitoring-based commissioning (continuous analytics)	1	4	1	4	1	**1.90**
2.4	Kitchen hood demand-controlled ventilation	1	2	2	2	2	**1.60**
2.5	Utility bill management & tariff review	1	2	1	2	1	**1.30**
2.6	Student & staff energy teams	1	4	1	3	1	**1.80**
3.1	LED fixtures with networked controls (fluorescent sites)	4	3	3	1	2	**3.10**
3.2	Exterior & parking-lot LED	3	2	1	1	1	**2.00**
3.3	Athletic field LED	4	3	2	1	3	**3.00**
4.1	Switchgear & distribution modernization	3	4	6	2	6	**4.00**
4.2	EHPA shelter generator upsizing	3	3	2	3	4	**2.90**
4.3	Rooftop solar PV (shelter sites)	4	3	1	3	4	**3.10**
4.4	Battery storage & islanding for shelters	4	4	1	4	6	**3.60**
4.5	Electric school bus charging	4	5	2	5	6	**4.10**
5.1	Roof replacement with insulation upgrade	10	4	4	1	5	**6.20**
5.2	Impact-rated window replacement	8	4	5	1	5	**5.60**
5.3	Exterior wall restoration & sealants	3	3	2	1	3	**2.60**
6.1	Fixture retrofits & smart irrigation	3	2	1	1	1	**2.00**
6.2	Domestic water piping replacement	5	4	7	1	6	**4.90**
6.3	Water submetering & leak detection	2	2	1	3	1	**1.80**
7.1	Elevate or relocate critical electrical	5	5	5	1	7	**4.80**
7.2	Storm drainage & backflow prevention	2	3	2	2	4	**2.40**
7.3	Wind-rated rooftop equipment anchorage	2	2	1	1	2	**1.70**
8.1	Access control & secure entry vestibules	5	4	3	3	4	**4.10**
8.2	Fire alarm & mass notification modernization	5	3	3	2	3	**3.70**
8.3	Camera system upgrade	3	3	1	3	2	**2.50**

Scores 1–10; higher = more costly. † Capital is not a workshop score: it is calculated from each measure's total project cost in fixed dollar bands (Section 10.2). Cost score = weighted sum of the five columns.

FIGURE 23

Workshop scorecard: benefit criteria

FIM	Measure	Utility savings (20%)	O&M & reliability (25%)	Learning env. (25%)	Operability (10%)	Safety & code (20%)	Benefit score	Quadrant
1.1	Packaged rooftop unit replacement (coastal-coated)	6	9	8	6	4	**6.85**	Quick win
1.2	Chiller plant replacement & optimization	8	10	8	7	4	**7.60**	Strategic investment
1.3	Cooling tower refurbishment	3	7	4	5	3	**4.45**	Opportunistic
1.4	Dedicated outdoor-air (DOAS) dehumidification	2	5	10	7	7	**6.25**	Strategic investment
1.5	Air-handler replacement with VAV conversion	6	8	8	6	4	**6.60**	Strategic investment
1.6	Heat-pump water heaters	5	5	2	3	2	**3.45**	Opportunistic
2.1	Building automation system upgrade	7	7	6	8	3	**6.05**	Quick win
2.2	Retro-commissioning	7	5	6	6	2	**5.15**	Quick win
2.3	Monitoring-based commissioning (continuous analytics)	4	6	4	7	2	**4.40**	Opportunistic
2.4	Kitchen hood demand-controlled ventilation	5	3	3	5	3	**3.60**	Opportunistic
2.5	Utility bill management & tariff review	4	2	1	6	1	**2.35**	Opportunistic
2.6	Student & staff energy teams	3	1	5	3	1	**2.60**	Opportunistic
3.1	LED fixtures with networked controls (fluorescent sites)	7	6	7	6	4	**6.05**	Quick win
3.2	Exterior & parking-lot LED	4	5	2	4	7	**4.35**	Opportunistic
3.3	Athletic field LED	5	6	5	4	4	**4.95**	Opportunistic
4.1	Switchgear & distribution modernization	1	8	3	5	8	**5.05**	Quick win
4.2	EHPA shelter generator upsizing	1	5	4	4	10	**4.85**	Opportunistic
4.3	Rooftop solar PV (shelter sites)	5	2	3	2	4	**3.25**	Opportunistic
4.4	Battery storage & islanding for shelters	3	2	3	3	9	**3.95**	Opportunistic
4.5	Electric school bus charging	6	4	2	4	3	**3.70**	Opportunistic
5.1	Roof replacement with insulation upgrade	2	9	7	6	8	**6.60**	Strategic investment
5.2	Impact-rated window replacement	3	6	7	4	9	**6.05**	Strategic investment
5.3	Exterior wall restoration & sealants	2	6	4	4	6	**4.50**	Opportunistic
6.1	Fixture retrofits & smart irrigation	5	3	2	4	2	**3.05**	Opportunistic
6.2	Domestic water piping replacement	1	8	6	5	7	**5.60**	Quick win
6.3	Water submetering & leak detection	4	4	1	5	2	**2.95**	Opportunistic
7.1	Elevate or relocate critical electrical	1	6	4	5	10	**5.20**	Quick win
7.2	Storm drainage & backflow prevention	1	6	4	4	8	**4.70**	Opportunistic
7.3	Wind-rated rooftop equipment anchorage	1	5	2	3	8	**3.85**	Opportunistic
8.1	Access control & secure entry vestibules	1	2	7	7	10	**5.15**	Quick win
8.2	Fire alarm & mass notification modernization	1	6	4	6	10	**5.30**	Quick win
8.3	Camera system upgrade	1	3	4	6	8	**4.15**	Opportunistic

Scores 1–10; higher = more beneficial. Benefit score = weighted sum of the five columns. Source: workbook Cost-Benefit sheet, rows 6–37.